



Impact of social cognitive propensity on the processing of nontransparent sentential meaning

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ARTICLE INFO

Article history:

Received 2 March 2022

Received in revised form 16 December 2022

Accepted 20 December 2022

Available online 13 January 2023

Keywords:

Nontransparent meaning

Meaning contextualization

Combinatorial semantic processing

Individual differences

Social cognition

Autism-spectrum quotient

ABSTRACT

This study investigates the influence of individual social-cognitive propensity in the processing of nontransparent sentential meaning, exemplified by the morpho-syntactically unsupported iterative meaning in “*The frog hopped for five minutes.*” Results of our speeded questionnaire in Mandarin Chinese showed that social cognitive propensity of typically-developed individuals, indexed by autistic-like traits, significantly correlated with online response times (RTs) of naturalness rating, while the effect was absent in offline rating scores. Individuals with higher autistic-like traits (i.e., lower social skills) took longer to process sentences with nontransparent meaning for making judgments. We argue that the computation of these sentences involves meaning contextualization—construing a coherent conceptual representation by integrating multiple lexical representations and evaluating sentential-discourse context. Such context-dependent meaning processing requires sufficient context sensitivity, which varies across individuals in association with social cognitive propensity. The pattern is captured by the Dual-Process Approach to information processing and social cognition: individuals with higher autistic-like traits are prone to deliberative reasoning with lower contextual sensitivity. This cognitive bias leads to greater cost when the full comprehension demands meaning contextualization, and therefore longer RTs in evaluating appropriate interpretations. The findings show that individual variability in social cognitive propensity modulates the online computation of nontransparent sentential meaning.

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1. Introduction

Meaning composition is regarded as the core linguistic computation (e.g., Martin and Baggio, 2020; Mollica et al., 2020). While the reading of a sentence largely comes from the meaning of the lexical items it contains and the way they are syntactically combined (Frege, 1892; Partee et al., 1990), very often the full, detailed interpretation goes beyond what is explicitly given by morphosyntactic cues. Here we examine an instantiation, exemplified in (1).

- (1) The frog hopped for 10 minutes in the backyard. → Multiple, iterative hopping

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The sentence engenders an iterative meaning that is not explicitly expressed by overt morphosyntactic means, which we term “nontransparent iterative meaning.” Notice that this nontransparent meaning emerges at the *combinatorial* (phrasal or sentential) level and not at the lexical level (not available from any single word). The phenomenon raises the questions of how the nontransparent iteration comes about, and what linguistic and cognitive factors contribute to such meaning comprehension. The issues can be addressed from two aspects: (a) linguistic devices that are given by input (external to comprehenders), and (b) comprehender-related, non-linguistic cognitive components that bear individual variability. While previous studies identify the linguistic factors that generate nontransparent iterative meaning (e.g., Deo and Piñango, 2011; Lai and Lai, 2022), here we probe the non-linguistic cognitive factors that may influence its processing, examining the impact of comprehenders’ social-cognitive propensity specifically.

We hold that the comprehension of nontransparent meaning involves a search in the sentential-discourse context (detailed in Section 1.1). Given the explicit linguistic cues, a full-blown interpretation is construed by the comprehender, making reference to the available conceptual and contextual information in a particular situation, along with one’s background knowledge. Take (1) for example; the comprehender obtains the iterative reading by taking sentential context into consideration, including both (a) the conventional length of the verbal predicate (*hop*) and (b) the interval length denoted by the temporal adverbial (*10 minutes*). In comparison, the counterparts “*The frog napped for 10 minutes*” with a different verb and “*The frog napped for 2 months*” with a different interval length denoted by the adverbial in addition, yield a continuous reading and an iterative reading, respectively. Juxtaposing these examples, one can see that the involvement of nontransparent iteration is dependent on sentential context.¹

The capacity of meaning-mining through contextual search varies across individuals however. Given that nontransparent meaning is computed by a context-dependent search on our hypothesis, *context sensitivity of comprehenders* factors into comprehension. To speak more generally, language is largely underspecified in the sense that not every piece of meaning is spelt out (e.g., Coulson and Oakley, 2005; Dowty, 2007; Frisson, 2009; Jackendoff, 1990, 1997; Pustejovsky, 1995; Ferreira et al., 2002); how well the comprehender makes use of contextual information is therefore crucial to achieve full comprehension. This suggests that individuals’ non-linguistic cognitive properties likely participate in language processing, in addition to linguistic devices, and could be partially the underlying sources of individual differences observed in psycholinguistic studies (e.g., Boland et al., 2016; Tager-Flusberg et al., 2005; Kidd et al., 2018). Recent findings on typically-developed individuals (TDs) support this view, showing that sentence comprehension is modulated by individual variability in context sensitivity, which is correlated with individuals’ autistic-like traits as a measure of social cognitive propensity (Lai et al., 2019; Zhang et al., 2022). This indicates that social cognition is recruited in language processing, consistent with the view of cognitive linguists that language is grounded in social interaction (e.g., Gibbs, 1994; Langacker, 2000, 2008; Tomasello, 2010).

In this study, we address the question of how social cognitive propensity associated with context sensitivity influences the processing of nontransparent combinatorial meaning in TDs. We hypothesize that individual social-cognitive propensity characterizes individual context sensitivity in information processing, and modulates the online computation of nontransparent meaning. Because its comprehension requires a context-dependent meaning search to evaluate event iterativity (detailed in Section 1.1), individual differences in context sensitivity would lead to differential processing profiles.² This predicts a correlation between individual social-cognitive propensity and real-time processing load, such that people with lower social-cognitive propensity, characterizing weaker context-sensitivity, would need more effort computing nontransparent meaning.

Individual differences in context sensitivity and the efficiency of exploiting contextual information have been revealed in two lines of research: studies on Autism Spectrum Disorder (ASD) and individual differences in TDs (e.g., Bendtz et al., 2022; Farmer et al., 2012; Kidd et al., 2018; Kozhevnikov et al., 2014). Autistic individuals with lower social skills encounter greater difficulty comprehending figurative language, which requires timely contextual evaluation and pragmatic inferences. The atypical language profiles are especially prominent in cases involving contextual integration, such as irony (e.g., Au-Yeung et al., 2015; Spotorno et al., 2012; Wang et al., 2006), metaphors (e.g., Kalandadze et al., 2018; Kasirer and Mashal, 2016; Rundblad and Annaz, 2010), and humor (e.g., Aykan and Nalçacı, 2018; Emerich et al., 2003; Lyons and Fitzgerald, 2004). The divergent processing profiles between ASDs and TDs likely result from differential cognitive styles in information processing (e.g., Ashwin and Brosnan, 2019; Brosnan et al., 2014), where *cognitive style* refers to how individuals process information, including perception, reasoning, problem-solving, decision-making, learning, and the way people relate to others. Importantly, individual differences are *graded* among the population rather than dichotomously determined by clinical diagnosis. All people within and across groups (TD and ASD) vary in the preferred way to handle information (e.g., Evans, 2003; Kozhevnikov, 2007; Kozhevnikov et al., 2014). The graded continuum of cognitive style is accompanied by the fact that autistic-like traits span across TDs (e.g., Baron-Cohen et al., 2001; Lewton et al., 2019; Ruzich et al., 2015). Processing variability in autistic tendency is therefore extendable to TDs: TDs with higher autistic-like traits pattern more like the ASD group than those with lower autistic-like traits.

¹ In the current view, “context” encompasses both sentential and discourse context. In the examples used here, only sentential context, which includes all linguistic cues present in the sentence, is available.

² The term “processing profiles” refers to patterns of real-time processing manifested as differential computation cost, to be measured by reading time or response time, for example, to experimental tasks.

Such individual variability in information processing style has been captured by the Dual-Process Approach (DPA), applicable to ASDs and TDs alike. In essence, it states that regardless of clinical group, individuals with higher autistic-like traits rely more on deliberative reasoning over intuition, with less sensitivity to context; conversely, people with lower autistic-like traits rely more on intuition than deliberation, exhibiting greater context sensitivity (Ashwin and Brosnan, 2019; Brosnan et al., 2016, 2017; De Martino et al., 2008; Happé et al., 2017; Rozenkrantz et al., 2021; Pijnacker et al., 2010). We reason that variability in autistic-like traits could result in individual differences in language processing *within the TD group*. This said, we underscore that language comprehension involves not only social cognition but also other cognitive factors (e.g., working memory, attention, abstraction of experiences, etc.); individual differences in language capacity could manifest in various dimensions. We target the social cognitive dimension as one instantiation, noting that the full range of examination awaits to be investigated.

In what follows, we will briefly introduce previous linguistic approaches to the phenomenon of nontransparent iterative meaning in Section 1.1, and show that its comprehension requires a context-dependent computation. In Section 1.2, we argue that nontransparent meaning computation is subject to individual variability in context sensitivity associated with autistic-like traits. This manifests an impact of social cognitive propensity that characterizes different cognitive styles of information processing. In Section 1.3, we discuss the specifics of the current study and outline our hypothesis. Section 2 presents a timed questionnaire that verifies our proposal. To forecast, results reported in Section 3 show that TDs with higher autistic-like traits deliberated longer when processing nontransparent meaning. The findings suggest that lower social propensity is associated with weaker context sensitivity, and less efficient information integration leads to more effortful computation of context-dependent meaning. We attribute the pattern to the differential reliance on logical reasoning vs. intuition in TDs as captured by the DPA to information processing. The findings from different research fields—linguistics, cognitive psychology, and developmental psychiatry—are unifiable, discussed in Section 4. Lastly, we conclude the study and discuss further implications in Section 5.

1.1. Nontransparent sentential meaning

In example (1), repeated below, the composition of a punctual verb (denoting a point-like event) with a durative *for*-adverbial (denoting an interval) engenders an iterative reading that there are multiple hopping events occurring during the 10-minute interval.

(1) The frog hopped for 10 minutes in the backyard. → Multiple, iterative hopping

The iterative meaning is not encoded in the lexical semantic representation of any component word, nor does it arise from a particular syntactic structure. The ultimate reading is obtained *beyond* transparent composition. Example (2) has the identical syntactic structure as (1) yet no iterative reading emerges.

(2) The frog napped for 10 minutes in the backyard. → One-time continuous napping

The contrast demonstrates that the iterative meaning in (1) is morpho-syntactically absent, hence “nontransparent,” yet fully obtained by comprehenders in the sentential context.

Linguistically, sentences like (1) manifest a syntax-semantics misalignment: the exact semantic-conceptual representation involving iteration is not predicted by the composition solely based on syntactic structure. Jackendoff (1997: 94) terms such a case “*enriched composition*” as opposed to simple composition in which the combinatorial meaning is syntactically transparent. Enriched composition seems to violate Fregean compositionality, which argues that the meaning of an expression comes from the meaning of its parts and the way they are syntactically combined (Frege, 1892). In real-time processing, sentences with nontransparent iterative meaning engender additional cost in behavioral measurements, found cross-linguistically in English (Piñango et al., 1999, 2006; Todorova et al., 2000), German (Bott, 2008, 2010), and Japanese (Ishii and Ishikawa, 2014; Lai et al., 2019) as well as additional brain activity, as compared to the transparent counterparts (e.g., Downey, 2006; Paczynski et al., 2014; Piñango and Zurif, 2001). The findings are not homogeneous however. Pickering et al. (2006) found no extra cost for processing nontransparent iterative meaning. They argue that comprehenders do not commit to full aspectual interpretation; rather, an exact, enriched interpretation is left underspecified due to processing economy. The phenomenon instantiates “*semantic underspecification*” accordingly.

Several linguistic analyses have been proposed to capture the nontransparent iteration and the associated processing cost. The conventional *Mismatch-Repair approach* claims that combining a durative adverbial with a punctual verb (e.g., *hop*, *sneeze*) results in an aspectual mismatch, due to the postulated selectional restriction of durative adverbials. The mismatch triggers a meaning-shifting operation (i.e., coercion) to repair, according to which the point-like event is interpreted iteratively.

(3) *The frog [hopped]_{punctual} [for 10 minutes]_{durative} in the backyard.*

→ ITER-operation → Iterative meaning

The operator insertion can be implemented as semantic coercion (e.g., de Swart, 1998; Jackendoff, 1997:51; Moens and Steedman, 1988; Piñango et al., 1999, 2006; Smith, 1994) or as pragmatic enrichment (Dölling, 2003, 2014; Egg, 2005). Regardless, the variants of this approach share the assumption that the meaning shift is motivated by the presence of a mismatch. The additional cost observed in psycho/neurolinguistic studies results from the insertion of this ITER-operation during comprehension.

The Mismatch-Repair approach has been challenged by recent studies however. Deo and Piñango (2011) point out that an iterative meaning arises in the absence of mismatch as well. This is exemplified by (4) in which the composition of a durative verb and a durative adverbial engenders an iterative meaning.

(4) *The frog [napped]_{durative} [for 2 months]_{durative} in the backyard.*

→ Iterative meaning

This shows that the emergence of nontransparent iteration is *not* conditioned by the presence of mismatch, casting the assumption of the Mismatch-Repair approach into doubt. By far, it is unclear how this approach accounts for the iteration with a durative verb, or one would have to capture the meaning via a separate operation (e.g., Dölling's (2014) habitual coercion).³

Without postulating a mismatch in the [punctual verb + durative *for*-adverbial] configuration, Deo and Piñango (2011) argue that both the one-time continuous reading and the iterative reading are available; the ultimate reading is determined by sentential and discourse context. The comprehender computes an appropriate interpretation by taking the following components into consideration: (a) the conventional, lexically encoded duration of the event denoted by the verb, (b) the interval denoted by the temporal *for*-adverbial, and (c) the larger discourse context if available. On this analysis, a temporal *for*-adverbial introduces an interval whose length is denoted by the complement noun phrase (e.g., *10 minutes*). The interval is partitioned into a set of sub-intervals in two ways: infinitesimal or non-infinitesimal partition measure. The comprehender applies an infinitesimal partition measure by default, which is too small to be distinguishable, and the sentence yields a continuous reading. If the continuous reading turns out implausible upon contextual evaluation in a given situation (reality or a fiction scenario), the comprehender implements a non-infinitesimal partition measure instead, segmenting the interval into a set of sub-intervals, each constituting a single event denoted by the verb (e.g., jumping, napping). The exact partition measure that results in the ultimate interpretation is contextually determined by (a)–(c) above. This *Context-Dependence approach* has gained support from recent experimental findings, which show that all sentences involving nontransparent iteration—with or without the mismatch postulated by the Mismatch-Repair approach—engendered additional and comparable processing cost (Piñango et al., 2016; Lai et al., 2019).

In consideration of theoretical and empirical advantages, we adopt the Context-Dependence approach for the following reasons. First, it captures sentences with nontransparent iterative meaning in a unifying manner by the mechanism of context-dependent meaning search. The approach is in line with Ariel's (2002a) perspective that nontransparent meaning is inferentially-derived from context given the linguistically-coded meaning. As it obviates a phenomenon-specific operation, the Context-Dependence approach also echoes Ziegeler's (2007b) proposal that the conventional notion of coercion can be explained by pragmatic means. Second, this treatment correctly predicts that all sentences involving nontransparent iteration—with or without a mismatch—incur greater cost than the transparent counterparts. Third and crucially, by pivoting meaning computation on contextual evaluation, the Context-Dependence approach naturally incorporates individual variability that is attributable to context sensitivity. This predicts an influence of social cognitive propensity in context-dependent computation, which is required in the comprehension of nontransparent meaning. On the other hand, individual variability and contextual impact on language processing is not clearly implicated by the Mismatch-Repair approach.

The expected individual variability highlights the role of *comprehenders* in meaning construal. As Michaelis (2004) mentions, the coercion effect is triggered by the interpreter when s/he has to reconcile sentence reading (also see Ziegeler, 2007a, p.996). The specific meaning obtained depends on the comprehender's conceptualization of the situation denoted by the coded meaning, the integration with his/her background knowledge, and his/her evaluation of contextual information. We term the process of construing the fully-specified meaning in a given sentential-discourse context “meaning contextualization.”

There are debates regarding whether nontransparent iterative meaning results from a semantic operation or pragmatic inference with extralinguistic knowledge nonetheless. The Mismatch-Repair approach offers a semantic solution.⁴ In contrast, Ziegeler (2007a) contends that cases like “*jumped until dawn*” instantiates a [SINGLE EVENT FOR MULTIPLE EVENTS] metonymy, computed through pragmatic inferencing. Ziegeler also notes that iteration arises in the absence of a durative adverbial (e.g., “*The frog hopped on the grass.*”) while the single semelfactive reading remains available, suggesting that the implied iteration is cancellable (p.1017). Accordingly, the nontransparent iterative meaning is considered an inferential part of

³ Also see Ziegeler's (2007a, p.998) comments on the coercion account in discussing the case of progressive predication with stative verbs (e.g., “*I'm liking your explanation.*”).

⁴ Dölling (2003, 2014) proposes a two-stage solution, such that a coercion operator is introduced when there is an aspectual mismatch, followed by pragmatic inferencing that enriches the iterative meaning. To the extent that it assumes a type mismatch in the composition of a punctual verb with a durative adverbial and that the mismatch is resolved via coercion operations, we regard it as pertaining to the Mismatch-Repair approach.

the verb's meaning in certain environments, and may become “foregrounded” gradually as a result of conventionalization. Such iterative expressions could be highly conventionalized implicatures, and therefore enjoy less degree of defeasibility. We refrain from attributing the phenomenon in question to semantics or pragmatics, nor do we assume such a distinction *a priori*, acknowledging the problem of maintaining the dichotomy.⁵ In the case of nontransparent meaning, it is likely that both semantic and pragmatic aspects (as traditionally defined) are involved. On Deo and Piñango's (2011) account adopted here, the search of an appropriate partition measure is driven by the semantics of *for*-adverbials, followed by pragmatic constraints with contextual evaluation. Rather than classifying the phenomenon, we find it more useful to explicate its nature and to examine how comprehension is achieved while being modulated by social cognition for our purpose.

It should be mentioned that nontransparent iterative meaning could arise in various constructions, encompassing sentences with *for*-adverbials (discussed by Deo and Piñango, 2011) and *until*-adverbials (exemplified in Ziegeler (2007a)). The two constructions may not involve exactly the same mechanism as Deo & Piñango mention (also see Downey's (2006) study). In Mandarin Chinese (MC), sentences with nontransparent iterative meaning under investigation do not have a preposition. In this language then, the emergence of iteration might be diagonal to specific prepositions. If there is a difference in processing mechanisms between sentences with *for*-adverbials vs. those with *until*-adverbials in English, that difference does not necessarily apply to MC. In spite of constructional variation, the essential mechanism shared across cases is a context-dependent meaning search to obtain an appropriate reading, while the exact information being searched for likely differs case-by-case. We capitulate on this context-dependent meaning search and ask how it is influenced by comprehenders' context sensitivity. And as context sensitivity characterizes social cognitive propensity (see Section 1.2 below), such effect of individual variability would demonstrate an impact of social cognitive propensity on language comprehension.

In sum, nontransparent meaning is interpreted in the sentential-discourse context, construed by the comprehender. On the Context-Dependence approach adopted here, the comprehender obtains nontransparent iteration via a contextual search for an appropriate partition measure, in consideration of sentential context (the verbal predicate, the interval length denoted by the temporal adverbial) and discourse if available. People nonetheless differ in context sensitivity, which is characterized by social cognitive propensity. We therefore expect to observe individual variability in processing nontransparent meaning, which requires context-dependent computation, as driven by context sensitivity. To probe this, we examine how the processing profile is modulated by individual context sensitivity indexed by autistic-like traits.

1.2. Social cognitive propensity & language comprehension

The interaction between nonlinguistic, domain-general cognition and domain-specific linguistic computation in language processing has long been acknowledged. For example, working memory and cognitive control are two nonlinguistic functions shown to support language processing (e.g., Daneman and Merikle, 1996; Just and Carpenter, 1992; Thothathiri et al., 2018). This dynamic interaction is natural since “language is embedded in a mosaic of cognitive functions” (Baggio et al., 2012). In Martin and Baggio's (2020) mechanistic view to language, understanding meaning composition requires a shift to identifying the neurocognitive mechanisms responsible for combinatorial computation.

One non-linguistic component that closely ties to language is *social cognition*, defined as “the processing of stimuli relevant to understanding agents and their interactions” by Happé et al. (2017). Beaudoin and Beauchamp (2020) take social cognition as “a complex set of mental abilities underlying social stimulus perception, processing, interpretation, and response.” To Frith and Frith (2007), it is the sum of the processes that allow interpersonal interaction, which essentially depends on the exchange of signals; the most obvious case is human speech communication. The specific definition of social cognition may vary slightly across researchers, but the consensus is that its constructs include interactive information processing and the understanding of others in communication. To the extent that language processing in conversations involves interpersonal social activity, social cognition is expected to exert nontrivial impact on linguistic computation and development (e.g., Tager-Flusberg et al., 2005; Tomasello, 2010).

The impact of social cognition on language is most clearly manifested in Autism Spectrum Disorder (ASD), whose deficient social skills lead to impaired language comprehension. They are less sensitive to context modulation in general, instantiated by homograph/homonym priming (Henderson et al., 2011), and err more in temporal-aspectual use (Perkins et al., 2006), while lexical access is comparatively intact (Tager-Flusberg et al., 2005). The atypical language profiles in ASD emerge most saliently in cases involving pragmatic processing, such as indirect speech, metaphors, and irony, in correspondence with divergent neural responses (e.g., Groen et al., 2010; Just et al., 2004; Kalandadze et al., 2018). The difficulty likely results from

⁵ A related issue is whether nontransparent meaning is literal (typically computed by semantic means) or nonliteral (computed via pragmatic inferences), if one assumes a dichotomy. We find it difficult to place the current case at either pole, because the nontransparent iterative meaning does not conform to all criteria of either literal or nonliteral meaning. While the nontransparent iteration emerges beyond transparent composition and may thus seem nonliteral, the iterative meaning is fairly salient and resembles literal meaning on the saliency dimension. This illustrates a problem of the conventional literal vs. nonliteral classification as noted by Ariel (2002a): nonliteral meaning is sometimes salient like literal meaning (p.364). In fact, such a dichotomy has been challenged on both theoretical and empirical grounds (Ariel, 2002a, b; Coulson and Oakley, 2005). What counts as literal meaning—traditionally defined as straightforwardly compositional, context-invariant, conventional, automatic, indirectly derived from the coded meaning—could vary in different contexts. Coulson and Oakley (2005) further suggest that meaning “is radically underspecified by linguistic information, which needs to be supplemented by contextually driven inferences in cases of literal and nonliteral meaning alike” even in seemingly straightforward cases (p. 1512).

weaker information integration (e.g., Gold et al., 2010; Minshew and Goldstein, 1998), contextual insensitivity (e.g., De Martino et al., 2008; Jolliffe and Baron-Cohen, 1999; Pijnacker et al., 2010), or/and greater reliance on deliberative logical reasoning on the Dual-Process Approach (DPA, Ashwin and Brosnan, 2019; Brosnan et al., 2014, 2016; Rozenkrantz et al., 2021).

In addition to the clinical group, typically-developed individuals (TDs) with higher autistic-like traits show greater effort in language comprehension, especially when pragmatic inferences are involved. Notice that autistic traits are graded among all individuals instead of being a categorical demarcation between the clinical ASD group and TDs. We shall take autistic traits as representing a cognitive bias, i.e., the way people process information, rather than a deficit (Baron-Cohen and Belmonte, 2005; Happé and Frith, 2006). Individual differences in this social cognitive propensity can be measured via the Autism-Spectrum Quotient (AQ), applicable to TDs as well as ASDs (Baron-Cohen et al., 2001). Just like ASD forms a spectrum with substantial variability within the clinical group, the spectrum manifests in TDs. That is, autistic-like traits form a continuum ranging across groups, showing variability within TDs as well (Baron-Cohen et al., 2001; Lundström et al., 2012; Ruzich et al., 2015). Accordingly, we reason that differential profiles of language processing between TDs vs. ASDs extend to TDs with low vs. high autistic-like traits, a view that has been supported by recent studies (Brosnan et al., 2016; Lewton et al., 2019).

This individual variability in social cognitive propensity can be captured by the Dual-Process Approach (DPA). Happé et al. (2017) highlight that social cognition consists of an automatic, cognitive efficient process and a deliberative reasoning process. Individual social propensity reflects how the intuitive/heuristic vs. deliberative/rational routes are used to achieve a particular goal (p.258). This perspective is shared by several researchers (e.g., Ashwin and Brosnan, 2019; St. Evans, 2008; Filmer et al., 2019; Frith and Frith, 2007). It is useful to distinguish between *ability* vs. *propensity* here. Happé et al. indicate that ASDs perform like TDs in a laboratory setting yet encounter pragmatic difficulty in real-life, suggesting that they may have the *ability* to exert social cognition, albeit with a reduced *propensity* to do so in real conversations (also see Cage et al. (2013) and Vivanti (2015) for the dissociation). The ability-propensity distinction interacts with the two systems of the DPA, such that “differences in propensity reflect the degree to which system 1 processes are automatically engaged during social interaction, whereas ability reflects the degree to which rational deliberative social reasoning (system 2) can produce accurate results” (Happé et al., 2017, p.261).⁶

The tendency towards rule-based, local information processing over intuitive, holistic processing implicates differential sensitivity to context. Ashwin and Brosnan (2019) explicitly indicate that “the greater deliberation seen in autism and people with high autistic-like traits is best characterized as being ‘unbiased’, or less influenced by context” (p.20). Studies have reported that autistic individuals show a reduced framing effect bias in using contextual information towards decision-making (De Martino et al., 2008; Shah et al., 2016). The pattern corresponds to the Weak Central Coherence theory, which states that ASDs are weaker at integrating information in global context while being superior at local-detail perception (Frith, 1989; Happé, 1999; Happé and Frith, 2006). Weaker contextual integration incurs greater cost in context-dependent processing, and this likely underpins their language difficulty.

We argue that the patterns of information processing between ASDs vs. TDs extend to TDs with high vs. low autistic-like traits. Because nontransparent meaning requires a context-dependent computation, its processing would likely be modulated by autistic-like traits in TDs according to the above discussion. In this sense, individual social-cognitive propensity factors into combinatorial meaning processing. Previous studies on TDs have shown evidence supporting this proposal. For example, Yang et al. (2018) found that the processing of scalar implicature, instantiated by the quantifier “some,” was impacted by both contextual setting and individual socio-pragmatic abilities measured by AQ in TDs. Higher-AQ adults showed lower context sensitivity to QUD (Question Under Discussion) for interpreting the meaning of *some*—they were more likely to accept “some” (pragmatically implicating “not all”) when all *n* of *n* items is referred to. Nieuwland et al. (2010) also report that higher-AQ TDs adopted literal meaning preferentially and consistently. In comprehending scalar implicatures, higher-AQ individuals were less sensitive to the pragmatic infelicity of under-informative sentences (e.g., “Some people have lung” vs. the informative counterpart “Some people have pet.”), exhibiting reduced N400 (lacking infelicity detection) in their ERP experiment. The findings suggest that TDs with higher autistic-like traits are prone to focus on logical reasoning, with less pragmatic inferences. Recently, Zhang et al. (2022) directly show that AQ scores in TDs significantly correlated with context sensitivity, modulating the context-dependent comprehension of English *have*-sentences (e.g., *The [maple tree (Location)/tall man (Possession)] has the car.*). These findings suggest that TDs differ in how well they can integrate information in a contextually appropriate manner, and that this variability is characterizable by autistic-like traits measured by AQ.

In short, the relationship between social cognition and language is observable in TDs by measures of autistic-like traits. On the DPA, individual differences in social cognitive propensity are related to differential reliance on deliberative reasoning with more focus on local details vs. intuitive thinking concerning more of the global context. People with higher autistic-like traits are prone to logical reasoning with weaker context sensitivity (to put it differently, they rely less on intuition with less dependence on context). By hypothesis, higher-AQ TDs (indexing lower social-cognitive propensity) are more context-insensitive and would show greater cost in comprehending nontransparent meaning that requires context-dependent computation. We integrate these dimensions as shown in Fig. 1.

⁶ System 1 refers to the intuitive/heuristic route, and system 2 refers to the logical/deliberational route.

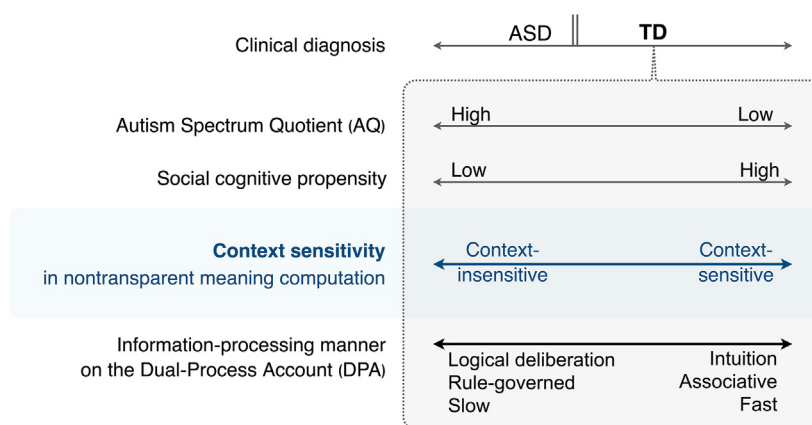


Fig. 1. Relationship of individual social-cognitive propensity indexed by AQ and context sensitivity involved during the computation of nontransparent meaning, as captured by the Dual-Process Account.

1.3. The current study

To summarize, the comprehension of nontransparent iterative meaning requires a context-dependent meaning search with conceptual-contextual integration, including sentential and extra-sentential information. Such meaning contextualization relies on the comprehender's context sensitivity, which is associated with social cognitive propensity indexable by AQ. Importantly, autistic-like traits that characterize context sensitivity is graded among TDs, the target population of the current study.

We hypothesize that individual variability in social cognitive propensity ties to variability in context sensitivity, reflecting different styles of information processing (logical deliberation vs. intuition) according to the DPA. The variability would modulate the processing profiles of nontransparent meaning in TDs. We predict a positive correlation between individual AQ scores and computation cost of processing nontransparent meaning. TDs with higher AQ scores, indexing lower social-cognitive propensity, are less context sensitive; therefore, they would need more effort to exert a context-dependent meaning search for the nontransparent meaning. The difficulty presumably results from more logical deliberation with lower efficiency of integrating sentential context information. Also, we expect to observe the effect of *propensity* in real-time processing, while offline measurements that index ultimate attainment more likely reflect the *ability* to obtain the meaning. These results would implicate a dynamic recruitment of social cognition in sentential meaning comprehension and provide a plausible explanation for individual differences long observed in language processing.

By far, most studies that examine the role of non-linguistic cognition in language processing target lexical (e.g., Gunter et al., 2003 on working memory; for social cognition, see Brock et al., 2008; Gaffrey et al., 2007; Lartseva et al., 2014; Kamio et al., 2007 for example) and syntactic parsing (e.g., Scheepers and Sturt, 2014; Thothathiri et al., 2018; Van de Cavey and Hartsuiker, 2016). Whether or not social cognitive propensity modulates *combinatorial meaning processing* in addition to typical figurative expressions (e.g., metaphors), and if this modulation would manifest in TDs are unclear. We aim to fill this gap by probing nontransparent sentential meaning. The phenomenon offers a valuable window to investigate how the impact of social cognitive propensity percolates to meaning processing beyond transparent composition, in the TD population specifically. By examining combinatorial meaning that is morphosyntactic-independent and requires sufficient context sensitivity, we can further clarify the specific aspect of the language system that is susceptible to social cognitive propensity.

2. Speeded questionnaire experiment

To examine how individual social-cognitive propensity influences nontransparent meaning comprehension, we implemented a speeded questionnaire in Mandarin Chinese (MC), in which the participants were asked to (i) rate the naturalness and (ii) determine an appropriate interpretation of sentences with time pressure. The materials and procedures were identical to Lai and Lai (2022); that study identifies the linguistic components (input given in sentences) that engender nontransparent iterative meaning in MC, including adverbial and verb types. The current study, on the other hand, examines the impact of one comprehender-related factor—context sensitivity characterized by social cognitive propensity—on meaning contextualization. We report novel results showing that individual variability in this aspect modulates the processing of nontransparent meaning.

2.1. Materials

We tested different conditions of sensical sentences to investigate whether or not the impact of individual social-cognitive propensity differed by linguistic configurations, or if this comprehender variability would emerge in a comparable manner across configurations. The latter was predicted more likely by our hypothesis: to obtain a full interpretation (the participants were asked to perform an interpretation task, see Section 2.3), the comprehender would implement a context-dependent search to determine the appropriate reading in the sentential context. We adopted thirty sets of four sensical conditions (A, B, C, D) from Lai and Lai (2022). The four differed in verb type and boundary specificity of the temporal adverbial (i.e., whether it specifies an exact interval length or not), as that study purports to identify the linguistic components contributing to the iterative reading in MC, a tenseless language. Condition-A included a post-verbal adverbial that encoded specific temporal boundaries (e.g., *20-miao* ‘twenty seconds’) in combination with a durative verb (e.g., *pao* ‘run’); this configuration yielded a transparent one-time continuous reading in the absence of further discourse context, serving as the baseline. The other three conditions involved nontransparent iterative meaning, which emerged in different configurations, including a punctual verb (e.g., *tiao* ‘jump’) combined with different adverbial types: with a post-verbal durative adverbial encoding specific temporal boundaries in Condition-B, or with a preverbal temporal adverbial without specific temporal boundaries (*buduandi/yizhi* ‘continuously’) in Condition-C. The two types of adverbials co-occurred with a punctual verb in Condition-D.

We further included twenty sets of three nonsensical conditions. In Condition-E, an achievement verb (e.g., *ying* ‘win’) was combined with a post-verbal durative adverbial. Condition-F included an individual-level stative verb (e.g., *gao* ‘be tall’) in composition with a post-verbal adverbial of the [for-x-time] or [in-x-time] type. Both conditions gave rise to aspectual incompatibilities. Condition-G manifested an animacy violation, in which an animate-selecting activity verb was paired with an inanimate-denoting subject. The nonsensical conditions were introduced for a few reasons. First, they balanced the experimental design given the naturalness-rating task, such that each response on the rating scale 1–5 (completely unnatural ~ completely natural) would be chosen. Second, these stimuli allowed examining if social cognitive propensity affected nonsensical and sensical sentences alike. The two were dissimilar in nature—a reasonable interpretation could be obtained (though with more effort) in sensical sentences, yet not in nonsensical ones; we were interested in whether or not the magnitude of individual differences would differ by sentence sensicality. A set of sample sentences was provided in Table 1. The full set of stimuli was provided in Appendix A.

On the Context-Dependence approach, comprehension of the sensical conditions involves an evaluation of the conventional length of the event denoted by the verb and the interval specified by the temporal adverbial. Assuming that this process of context-dependence computation would take place for all sensical sentences alike, we expected to observe similar effect of social cognitive propensity. With respect to nonsensical sentences, we remained open because the role of individual propensity in processing sentences that have no reasonable meaning is less known. Our results and Lai & Lai’s findings, when taken together, reveal both the linguistic and nonlinguistic individual factors that give rise to the real-time processing profiles of nontransparent iterative meaning in MC.

2.2. Participants

Fifty-six native speakers (32 females) of Taiwanese Mandarin Chinese (MC) were recruited via advertisement posting on campus. All participants had normal or corrected-to-normal vision, without reading disabilities by self-report, between 18–30 years old (mean age = 21.83). Written informed consent was obtained from each participant prior to the experiment.

2.3. Procedure

The participants took a timed sentence-comprehension questionnaire, followed by the Autism-Spectrum Quotient (AQ, Baron-Cohen et al., 2001) in its Chinese version (Liu, 2008). In the questionnaire, the participants rated the naturalness of each sentence from a scale 1–5 (5 = completely natural) as fast as possible upon full comprehension. After rating, they were asked to judge the iterativity of the event denoted by the sentence just read, among four answer options: “once”, “more than twice (not only once)”, “possibly once or multiple times”, and “nonsensical.” This interpretation task was introduced to confirm that an iterative reading was obtained for Condition B, C, D as expected.

The questionnaire was implemented via the E-Prime 3.0 software (Psychology Software Tools, Pittsburgh, PA). The rating scores and the response time (RT) were recorded as the dependent variables (DVs). A short practice session was given prior to the real trials to familiarize the participants with the procedures. During the real-trial session, they had a chance to take a brief break after every five sentences. Following the sentence-comprehension questionnaire, the participants took AQ that consisted of 50 statements. For each statement the participants judged whether it characterized them appropriately, by

Table 1
Conditions and sample sentences.

Stimuli Type	Condition VerbType_AdvType	Sample sentence
Sensical No Iteration	(A) Durative_Post	<i>Xiaopengyou zai chuang-shang ku-le 3 fenzhong,</i> little child LOC bed-top cry-LE 3 minute <i>bu-ken shanbaganxiu.</i> NEG-willing give up 'The little child cried for 3 minutes on the bed, not willing to give up.'
Sensical Nontransparent Iteration	(B) Punctual_Post	<i>Xiaopengyou zai chuang-shang tiao-le 3 fenzhong,</i> little child LOC bed-top jump-LE 3 minute <i>bu-ken shanbaganxiu.</i> NEG-willing give up 'The little child jumped for 3 minutes on the bed, not willing to give up.'
Sensical Nontransparent Iteration	(C) Punctual_Pre	<i>Xiaopengyou zai chuang-shang buduandi tiao,</i> little child LOC bed-top continuously jump <i>bu-ken shanbaganxiu.</i> NEG-willing give up 'The little child continuously jumped on the bed, not willing to give up.'
Sensical Nontransparent Iteration	(D) Punctual _Pre+Post	<i>Xiaopengyou zai chuang-shang buduandi tiao-le 3 fenzhong,</i> little child LOC bed-top continuously jump-LE 3 minute <i>bu-ken shanbaganxiu.</i> NEG-willing give up 'The little child continuously jumped for 3 minutes on the bed, not willing to give up.'
Nonsensical	(E) Achiev_DurAdv	<i>Yundongyuan zongjuesai ying-le 10 fenzhong,</i> athlete final game win-LE 10 minute <i>xiande hen deyi.</i> look very proud 'The athlete won the final game for 10 minutes and looked very proud.'
Nonsensical	(F) State_DurAdv	<i>Yundongyuan zongjuesai gao-le 10 fenzhong,</i> athlete final game tall-LE 10 minute <i>xiande hen deyi.</i> look very proud 'The athlete was tall for 10 minutes in the final game and looked very proud.'
Nonsensical	(G) SemViolation	<i>Jiaokeshu zai caochang pao-le yi-ge shangwu,</i> textbook LOC playground run-LE one-CL morning <i>xiande hen deyi.</i> look very proud 'The textbook ran on the playground all morning and looked very proud.'

selecting among four options: “*definitely agree*”, “*slightly agree*”, “*slightly disagree*” or “*definitely disagree*.” The experiment was approved by the Institutional Review Boards of National Chengchi University, Taiwan.

2.4. Data analysis

Two participants were identified as outliers (rating score SD > 2.5 by subject-analysis in data screening) and their data were excluded prior to further statistical analysis. Problematic trials due to technical issues were removed (0.258% of the data points). Following Schütze and Sprouse (2014), we transformed the rating scores into z-scores and excluded data points exceeding 2.5 SD from the means to avoid distortions from extreme values. The data-preprocessing step resulted in 5.37% data lost of the whole dataset (5.57% for sensical conditions and 4.98% for nonsensical conditions respectively).

All statistical analyses were performed in the R statistical environment (R Core Team, 2021). Given the disparate *a priori* manipulations and sensibility of the sentences, we analyzed the sensical and nonsensical conditions separately. For each set of conditions, mixed-effects model analysis was conducted to access the effect of AQ scores on the rating z-score and the rating-RTs respectively (Baayen et al., 2008), using the *lmer* function of the *lme4* package (Bates et al., 2015). We accessed effect significance by backward model comparisons via likelihood ratio tests (Baayen et al., 2008; Field et al., 2012; Winter, 2013),

contrasting a model with the fixed factor in question against a base model without it. All models incorporated random intercepts for subject and item-set as well as a random slope for condition over item-set, as more complicated models failed to converge.⁷ In addition, we performed correlation analyses to evaluate the relationship between individual AQ scores and the two DVs, rating scores and rating-RTs.⁸ The same procedure of data analysis was applied to sensical and nonsensical sentences.

3. Results

Descriptive statistic results of the naturalness-rating scores are summarized in Table 2, visualized in Fig. 2. All sensical conditions were rated within the natural range in average (>4.3), whereas the nonsensical sentences were rated much lower, below the mid-point (<2.1). Results of the interpretation task confirmed that the participants obtained the iterative reading for the 3 conditions assumed to involve the nontransparent iterative meaning unambiguously: Condition (B) Punctual_Post: 81.05%, Condition (C) Punctual_Pre: 89.94%, Condition (D) Punctual_Pre+Post: 87.96%. Condition (A) Durative_Post engendered the one-time continuous reading mostly (62.41%). These descriptive results of sensical conditions were reported in Lai and Lai (2022).

Table 2

Descriptive summary of naturalness rating scores.

Stimuli Type	Condition	Mean	SD	Median	se
Sensical	(A) Durative_Post	4.32	0.77	4	0.02
	(B) Punctual_Post	4.50	0.64	5	0.02
	(C) Punctual_Pre	4.60	0.63	5	0.02
	(D) Punctual_Pre+Post	4.42	0.70	5	0.02
Nonsensical	(E) Achiev_DurAdv	1.91	0.96	2	0.03
	(F) State_DurAdv	1.59	0.84	1	0.03
	(G) SemViolation	2.04	1.21	2	0.04

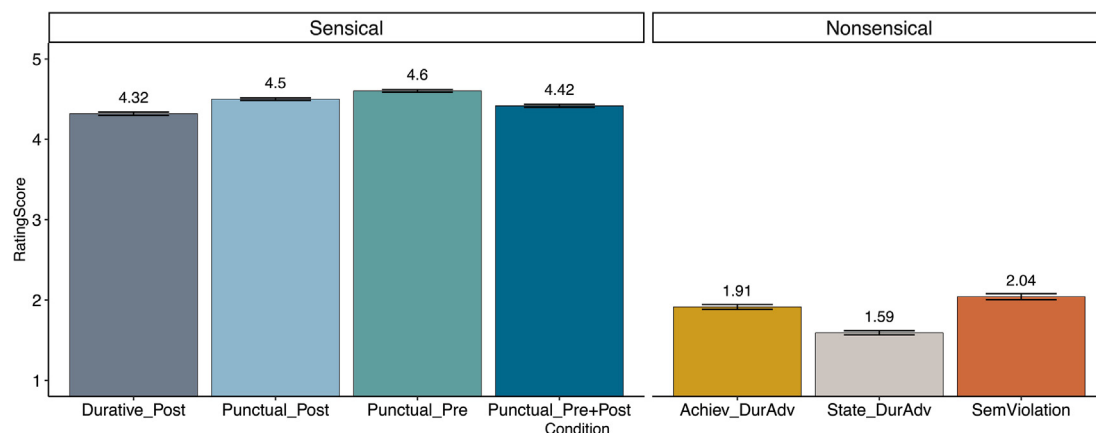


Fig. 2. Mean scores of naturalness rating per condition.

The unique investigation of this study is the effect of AQ on the processing of nontransparent meaning. The rating scores for sensical sentences revealed a significant main effect of Condition ($X^2(3) = 38.53, p < .001$). The AQ scores and its interaction with Condition were insignificant. For the nonsensical sentences, the rating scores showed a significant main effect of Condition ($X^2(2) = 25.76, p < .001$) and a significant AQ scores $\times$ Condition interaction ($X^2(2) = 18.92, p < .001$). No main effect of AQ scores was found.

Regarding Rating-RTs, results of the sensical sentences showed a significant main effect of AQ scores ($X^2(1) = 10.16, p = .001$), and a significant main effect of Condition ($X^2(3) = 36.73, p < .001$). No interaction between AQ and Condition was found ($X^2(3) = 4.72, p = .19$). The nonsensical sentences elicited a significant main effect of AQ scores on Rating-RTs ($X^2(1) = 4.80, p = .028$), and a significant main effect of Condition ($X^2(2) = 20.13, p < .001$). The interaction between Condition and AQ scores was significant as well ($X^2(2) = 8.28, p = .016$).

Results of *Pearson*-correlation coefficients showed a positive correlation between Rating-RTs and individual AQ scores, in sensical sentences taken as a group ($r = 0.42, p = .002$), visualized in Fig. 3a. For the sensical sentences, moderate positive

⁷ The maximal model incorporating the interaction of the two fixed factors for Rating-RTs was specified as: $lmer(RT \sim Condition * AQscore + (1|Subject) + (1+Condition|ItemSet), data = data, REML = F)$. The significance of the interaction effect was assessed by contrasting this maximal model with the additive model, where the fixed-effect structure was replaced by $[Condition + AQscore]$. The main effect AQ score and Condition were assessed by contrasting the additive model with the effect in question dropped respectively.

⁸ The *cor.test()* function and the *ggpubr* packages were used for computing the correlations.

correlations were found for each condition: Condition (A) Durative_Post: $r = 0.36$, Condition (B) Punctual_Post: $r = 0.46$, Condition (C) Punctual_Pre: $r = 0.38$, Condition (D) Punctual_Pre+Post: $r = 0.42$ (Fig. 3c). For the nonsensical conditions, the positive correlations between individual AQ scores and Rating-RTs were relatively weaker ($r = 0.29$, $p = .032$; Fig. 3b). The effect was more salient in Condition (E), the [Achievement + Durative Adv] construction ($r = 0.34$), and Condition (F), the [State + Durative Adv] construction ($r = 0.30$) as shown in Fig. 3d. The correlation for Condition (G) [Semantic Violation], turned out insignificant ($r = 0.17$). On the other hand, the correlations between offline rating scores and AQ scores were insignificant for sensical and nonsensical sentences, both at the condition-converged level and the single-condition level.

These results allow us to address if the impact of social cognitive propensity differed across linguistic configurations. The pattern suggests that while this individual factor modulated the Rating-RTs of both sensical and nonsensical conditions, the strength of impact—indicated by a positive AQ-RT correlation—was stronger and more consistent within the four sensical

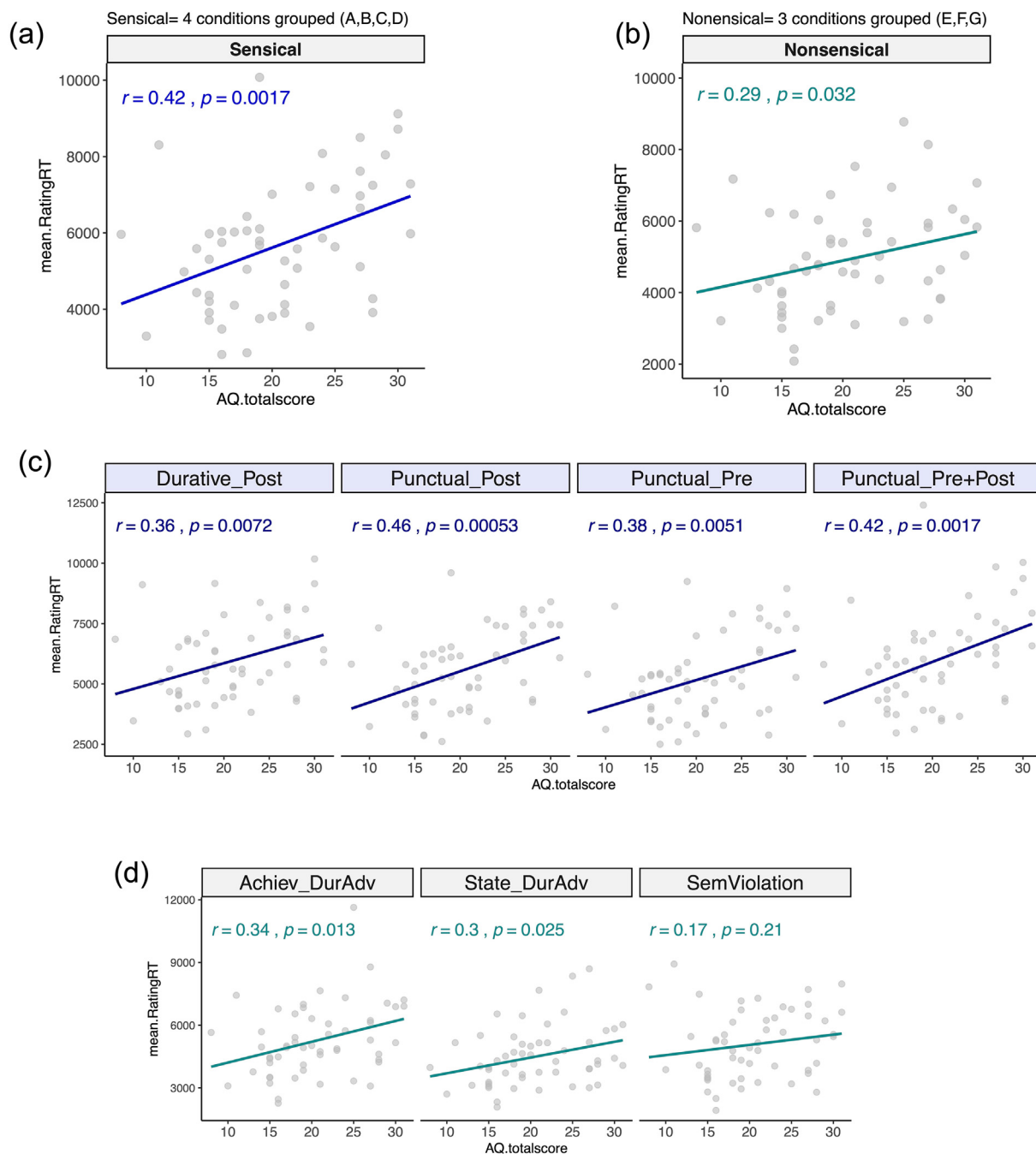


Fig. 3. Correlations between individual AQ scores and Rating-RT (in milliseconds).

conditions, and less so for the three nonsensical conditions. This can be observed by comparing Fig. 3c vs. Fig. 3d. Such variations did not emerge in the offline rating z-scores. More details with additional analyses are provided in footnote.⁹

In sum, individual AQ scores significantly correlated with online response times but not offline rating scores. The findings that individuals with higher vs. lower autistic traits exhibited differential profiles suggest that (i) social cognitive propensity modulates real-time comprehension, and (ii) individuals likely implement processing strategies differentially when computing nontransparent sentential meaning.

4. Discussion

Results of our questionnaire showed that the higher the autistic-like traits, the longer time it took to make the ultimate judgment for a given sentence. Crucially, significant individual variability emerged in online rating-RTs but not offline rating scores, indicating that the impact of social cognitive propensity took place during real-time computation of nontransparent meaning. The asymmetry between online vs. offline measurements suggests that although all participants arrived at the ultimate interpretation alike, the processing strategies and the cognitive mechanisms involved may differ across individuals. We argue that such individual variability in nontransparent meaning processing is capturable by social cognitive propensity with respect to context sensitivity.

On the Context-Dependence approach, comprehending nontransparent iterative meaning essentially requires the retrieval of lexical conceptual representations, integrating these representations in sentential context (the event denoted by the verb and the interval-length denoted by the temporal adverbial), and evaluating against the larger discourse context whenever available. Accordingly, lower context sensitivity or less efficient information integration would slow down the context-dependent meaning computation. Given that context sensitivity negatively correlated with autistic-like traits in TDs (Zhang et al., 2022), we predicted that higher-AQ TDs would need more time to compute nontransparent meaning. This is exactly what we observed in the RT results (Figs. 3 and 4).

The findings thus illustrate the impact of individual social-cognitive propensity in sentential meaning comprehension, probed via autistic-like traits that index context sensitivity. The pattern exhibited by high-AQ individuals likely reflects weaker efficiency in processing complex information, with greater reliance on deliberative reasoning over intuition, and with less contextual dependence by the Dual-Process Approach (DPA; Brosnan et al., 2016; De Martino et al., 2008; Lewton et al., 2019). These properties are more likely intercorrelated rather than mutually exclusive, along the mechanistic view to language. More specifically, on the DPA, individuals with higher autistic-like tendency show enhanced rationality, implementing a higher level of deliberative over intuitive reasoning in decision making (Brosnan et al., 2014, 2016) and syllogism deduction (Lewton et al., 2019). De Martino et al. (2008) report that autistic individuals were less susceptible to the context-framing effect (incorporating less contextual information into decision-making processes in their gambling task). These boil down to Ashwin and Brosnan's (2019) and Rozenkrantz et al. (2021) DPA for autism. We extend this account to TDs with higher vs. lower autistic-like traits to capture individual differences in real-time language comprehension, which is essentially a kind of information processing. This interpretation echoes the hyper-systemizing theory: the tendency of autistic individuals towards hyper-attention or hyper-sensitivity to low-level, rule-based input is accompanied by diminished or optional higher-order computation (Baron-Cohen et al., 2009; Greenberg et al., 2018; Motttron et al., 2006). Accordingly, the longer Rating-RTs in higher-AQ TDs found in our experiment could result from more deliberation on inferring the nontransparent meaning.

In other words, context sensitivity, complex information integration, and the DPA to different styles of information processing might be intercorrelated in contributing to the observed pattern. The lower an individual's social propensity (higher AQ), the weaker ability to take various cues into consideration simultaneously, due to hyper-attention to local/low-level input, resulting in less efficient information integration. Yet, meaning contextualization by integrating multiple lexical representations with a given sentential-discourse context is indispensable to obtain an appropriate interpretation for sentences with nontransparent meaning. A weaker efficiency of context integration thus leads to more effortful meaning contextualization, engendering greater processing effort, manifested as longer RTs in higher-AQ individuals.

In short, the additional processing load of nontransparent sentential meaning is attributed to meaning indeterminacy on the Context-Dependence approach. Computing a contextually appropriate interpretation requires more effort when the

⁹ We performed two additional sets of mixed-effects model analysis with the same model comparison procedure above. First, we incorporated a fixed factor of *Sensicality* with 2 levels of stimuli type (sensical, nonsensical) in interaction with AQ scores in the model, specified as below (the random-effect structure was simplified from the more complicated ones that failed to converge): $\text{lmer}(\text{DV} \sim \text{Sensicality} \times \text{AQscore} + (1 + \text{Sensicality} | \text{Subject}) + (1 | \text{ItemSet}), \text{data} = \text{data}, \text{REML} = \text{F})$. Results of rating z-scores showed a significant effect of *Sensicality* ($\chi^2(1) = 8.92, p < .003$), with no main effect or AQ nor *Sensicality* $\times$ AQ interaction. This indicates that all participants rated the sentences by sensicality in a similar manner. In comparison, results of Rating-RTs showed a significant effect of AQ ($\chi^2(1) = 8.82, p < .003$; higher AQ > lower AQ in RTs) and of *Sensicality* ($\chi^2(1) = 8.75, p = .003$; sensical > nonsensical in RTs); the *Sensicality* $\times$ AQ interaction was insignificant. This corroborates the findings that online computation varied with individual social-cognitive propensity, and such variability (a positive correlation between RTs and AQ scores) emerged in both stimuli types. Second, we incorporated a fixed factor of *Condition* with 7 levels (Condition A–G), along with AQ scores in the model, specified as below: $\text{lmer}(\text{DV} \sim \text{Condition} \times \text{AQscore} + (1 | \text{Subject}) + (1 | \text{ItemSet}), \text{data} = \text{data}, \text{REML} = \text{F})$. Results of rating z-scores showed a significant effect of *Condition* ($\chi^2(6) = 347.53, p < .001$), and a significant *Condition* $\times$ AQ interaction ($\chi^2(6) = 26.64, p < .001$); the main effect of AQ was insignificant. Further analyses revealed that the correlations between rating scores and AQ differed among the conditions, yet none of the conditions showed a significant effect in particular. With respect to Rating-RTs, we found a significant effect of *Condition* ($\chi^2(6) = 135.26, p < .001$; see Lai and Lai (2022) discussion on the sensical conditions), an effect of AQ ($\chi^2(1) = 10.55, p = .001$), and a significant *Condition* $\times$ AQ interaction ($\chi^2(6) = 30.03, p < .001$). Taken these findings together with the correlation results, the pattern suggests that individual social-cognitive propensity modulated Rating-RTs, with differential strengths of impact across conditions (i.e., the positive correlations were stronger for sensical conditions than for nonsensical conditions). We thank an anonymous reviewer for suggesting these analyses.

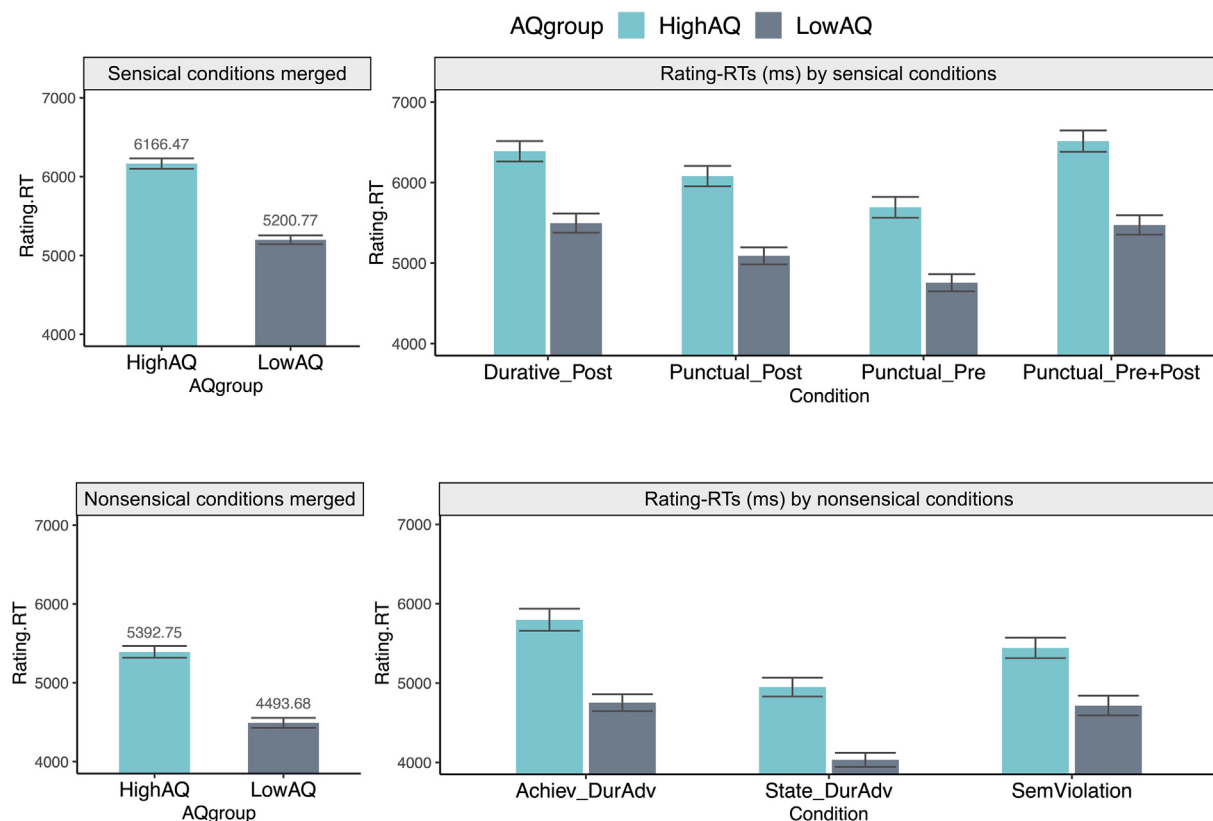


Fig. 4. Bar charts showing Rating-RTs in milliseconds (mean $\pm$ 1 standard errors) by High vs. Low AQ group (2-tiled, each having 27 participants). Descriptive statistics of AQ scores per group: Mean_{High-AQ} = 25.26, SD_{High-AQ} = 3.49, Median_{High-AQ} = 25; Mean_{Low-AQ} = 15.74, SD_{Low-AQ} = 2.8, Median_{Low-AQ} = 16.

given/coded linguistic input is insufficient for determining an exact, full-blown interpretation. This is particularly difficult when (i) the comprehender has weaker context sensitivity and less integrative consideration, or when (ii) a guiding context is absent. Our account is consistent with [Lai and Piñango's \(2019\)](#) findings, which showed that the more ambiguous a sentence remains in a non-informative context (vs. biasing context), the lower comprehensibility it appears to readers. This speaks to degrees of *informativity* or *meaning specificity*: the less information is given explicitly by linguistic devices, the more the comprehender needs to search so as to contextualize sentence meaning, translatable to greater processing load. Consequently, individual variability of meaning contextualization efficiency gives rise to variability in real-time processing profiles.

5. Conclusion & implications

This study shows that online processing of nontransparent sentential meaning is modulated by social cognitive propensity as indexed by individual autistic-like traits. Typically-developed individuals with higher AQ require more time computing nontransparent sentential meaning than those with lower AQ. We account for variability in language processing by individual differences in context sensitivity and efficiency of information integration. This pattern is capturable by the Dual-Process Account as differential cognitive styles—deliberative reasoning vs. intuition in information processing and decision-making. We argue that meaning contextualization—construing an appropriate meaning in context by the comprehender—is modulated by the ability of context integration that underpins social cognitive propensity, measurable by autistic-like traits. The findings offer novel evidence that implicates a significant role of social cognition in the real-time computation of combinatorial meaning. While the interaction of the two has been acknowledged, the current study presents a concrete instantiation showing that social cognition supports the online processing of nontransparent combinatorial meaning independent of morphosyntactic components. Such an effect in a typologically different and tenseless language, Mandarin Chinese, further corroborates previous studies that are mainly done in Indo-European languages, revealing cross-linguistic commonalities.

Situating these findings in a larger scope, we speculate that the results provide clues for evaluating the emergentist approaches (e.g., the usage-based approach or constraint-based approaches) and the formal linguistic approach (in which the subsystem of language is considered relatively vertical, independent of use) to language. The Context-Dependence account to combinatorial meaning is close to the former perspective, which naturally incorporates a strong interaction between linguistic and other cognitive systems (*cf.* [Martin and Baggio, 2020](#)). The relationship is observable via individual variability in

social cognitive propensity on language processing in TDs. Future research is called for to examine if such impact of individual cognitive style applies to other cases of nontransparent meaning alike (e.g., other coercion phenomena, various types of metonymy and metaphors etc.), or if it turns out graded across cases.

Declaration of competing interest

There is no conflict of interest for this work.

Data availability

Data will be made available on request.

Acknowledgement

This work is supported by the National Science and Technology Council, NSTC (previously named Ministry of Science and Technology, MOST), Taiwan, Grant# MOST 109-2410-H-004-192-MY2 & MOST 111-2410-H-004-097 to the first author, and Grant# MOST 111-2410-H-004-089-MY2 to the second author. We thank Xiang-Ru Chen, Yu-Ju Chiu and Chien-Yu Lin (in alphabetical order) for helping data collection. We would like to express our gratitude to the two anonymous reviewers for their constructive comments, and to the Co-Editor in Chief, Anne Bezuidenhout, for her kind assistance on this manuscript.

Appendix A

Appendix A-1. Stimuli – Sensical sentences

Set#	Condition	Target Sentence
s01	(A) Durative_Post (No Iteration)	Yundongyuan pao-le 20 miao, yidian dou bu jue de lei. athlete run-LE 20 second little at all NEG feel tired 'The athlete ran for 20 seconds, and did not feel tired at all.'
	(B) Punctual_Post (Nontransparent Iteration)	Yundongyuan tiao-le 20 miao, yidian dou bu jue de lei. athlete jump-LE 20 second little at all NEG feel tired 'The athlete jumped for 20 seconds, and did not feel tired at all.'
	(C) Punctual_Pre (Nontransparent Iteration)	Yundongyuan buduandi tiao, yidian dou bu jue de lei. athlete continuously jump little at all NEG feel tired 'The athlete continuously jumped, and did not feel tired at all.'
	(D) Punctual_Pre+Post (Nontransparent Iteration)	Yundongyuan buduandi tiao-le 20 miao, yidian dou bu jue de lei athlete continuously jump-LE 20 second little at all NEG feel tired 'The athlete continuously jumped for 20 seconds, and did not feel tired at all.'
s02	(A) Durative_Post (No Iteration)	Laoyeye zai zhen-jian tang-le 1 fenzhong, sihu hen bushufu. elderly man LOC clinic lie-LE 1 minute seem very uncomfortable 'The elderly man lay for 1 minute in the clinic and seemed very uncomfortable.'
	(B) Punctual_Post (Nontransparent Iteration)	Laoyeye zai zhen-jian ke-le 1 fenzhong, sihu hen bushufu. elderly man LOC clinic cough-LE 1 minute seem very uncomfortable 'The elderly man coughed for 1 minute in the clinic and seemed very uncomfortable.'
	(C) Punctual_Pre (Nontransparent Iteration)	Laoyeye zai zhen-jian yizhi ke, sihu hen bushufu. elderly man LOC clinic continuously cough seem very uncomfortable 'The elderly man continuously coughed in the clinic and seemed very uncomfortable.'
	(D) Punctual_Pre+Post (Nontransparent Iteration)	Laoyeye zai zhen-jian yizhi ke-le 1 fenzhong, elderly man LOC clinic continuously cough-LE 1 minute sihu hen bushufu. seem very uncomfortable 'The elderly man continuously coughed for 1 minute in the clinic and seemed very uncomfortable.'
s03	(A) Durative_Post (No Iteration)	Zhanzheng youxi li zhanshi gong-le 20 miao, haishi mei dabai dijun. war game in warrior attack-LE 20 second still NEG defeat enemy 'The warrior attacked for 20 seconds in the war game and still did not defeat the enemy.'
	(B) Punctual_Post (Nontransparent Iteration)	Zhanzheng youxi li zhanshi ti-le 20 miao, haishi mei dabai dijun. war game in warrior kick-LE 20 second still NEG defeat enemy

		'The warrior kicked for 20 seconds in the war game and still did not defeat the enemy.'
	(C) Punctual_Pre (Nontransparent Iteration)	Zhanzheng youxi li zhanshi buduandi ti, haishi mei dabai dijun. war game in warrior continuously kick still NEG defeat enemy 'The warrior continuously kicked in the war game and still did not defeat the enemy.'
	(D) Punctual_Pre+Post (Nontransparent Iteration)	Zhanzheng youxi li zhanshi buduandi ti-le 20 miao, war game in warrior continuously kick-LE 20 second haishi mei dabai dijun. still NEG defeat enemy 'The warrior continuously kicked for 20 seconds in the war game and still did not defeat the enemy.'
s04	(A) Durative_Post (No Iteration)	Xiao nuhai zai zhuozi-shang pa-le 20 miao, juede hen wuliao. little girl LOC table-top lie-LE 20 second feel very bored 'The girl lay on the table for 20 seconds, feeling very bored.'
	(B) Punctual_Post (Nontransparent Iteration)	Xiao nuhai zai zhuozi-shang deng-le 20 miao, juede hen wuliao. little girl LOC table-top pedal-LE 20 second feel very bored 'The girl pedaled on the table for 20 seconds, feeling very bored.'
	(C) Punctual_Pre (Nontransparent Iteration)	Xiao nuhai zai zhuozi-shang yizhi deng, juede hen wuliao. little girl LOC table-top continuously pedal feel very bored 'The girl continuously pedaled on the table, feeling very bored.'
	(D) Punctual_Pre+Post (Nontransparent Iteration)	Xiao nuhai zai zhuozi-shang yizhi deng-le 20 miao, little girl LOC table-top continuously pedal-LE 20 second juede hen wuliao. feel very bored 'The girl continuously pedaled on the table for 20 seconds, feeling very bored.'
s05	(A) Durative_Post (No Iteration)	Lujun budui zai yuandi dun-le 10 fenzhong, bu-zhi heshi cai neng xiuxi. army force LOC in place squat-LE 10 minute, NEG-know when CAI can rest 'The army squatted in place for 10 minutes, not knowing when they can rest.'
	(B) Punctual_Post (Nontransparent Iteration)	Lujun budui zai yuandi ta-le 10 fenzhong, bu-zhi heshi cai neng xiuxi. army force LOC in place step-LE 10 minute, NEG-know when CAI can rest 'The army stepped in place for 10 minutes, not knowing when they can rest.'
	(C) Punctual_Pre (Nontransparent Iteration)	Lujun budui zai yuandi buduandi ta, bu-zhi heshi cai neng xiuxi. army force LOC in place continuously step, NEG-know when CAI can rest 'The army continuously stepped in place, not knowing when they can rest.'
	(D) Punctual_Pre+Post (Nontransparent Iteration)	Lujun budui zai yuandi buduandi ta-le 10 fenzhong, army force in place continuously step-LE 10 minute, bu-zhi heshi cai neng xiuxi. NEG-know when CAI can rest 'The army continuously stepped in place for 10 minutes, not knowing when they can rest.'
s06	(A) Durative_Post (No Iteration)	Chushi yong rou-chui ya-le 10 miao, cai ba rou da-bian. cook use meat-hammer press-LE 10 second CAI BA meat beat-flat 'The chef pressed for 10 seconds with a meat hammer before flattening the meat.'
	(B) Punctual_Post (Nontransparent Iteration)	Chushi yong rou-chui pai-le 10 miao, cai ba rou da-bian. cook use meat-hammer pat-LE 10 second CAI BA meat beat-flat 'The chef patted for 10 seconds with a meat hammer before flattening the meat.'

	(C) Punctual_Pre (Nontransparent Iteration)	Chushi yong rou-chui buduandi pai, cai ba rou da-bian. cook use meat-hammer continuously pat CAI BA mean beat-flat 'The chef continuously patted with a meat hammer before flattening the meat.'
	(D) Punctual_Pre+Post (Nontransparent Iteration)	Chushi yong rou-chui buduandi pai-le 10 miao, cook use meat-hammer continuously pat-LE 10 second cai ba rou da-bian. CAI BA meat beat-flat 'The chef continuously patted for 10 seconds with a meat hammer before flattening the meat.'
s07	(A) Durative_Post (No Iteration)	Zhuang-han yong quantou yal-e 1 fenzhong, hai bu jue de tong. strong-man use fist press-LE 1 minute still NEG feel pain 'The strong man pressed for a minute with his fists, but didn't feel pain.'
	(B) Punctual_Post (Nontransparent Iteration)	Zhuang-han yong quantou chui-le 1 fenzhong, hai bu jue de tong. strong-man use fist beat-LE 1 minute still NEG feel pain 'The strong man beat for a minute with his fists, but didn't feel pain.'
	(C) Punctual_Pre (Nontransparent Iteration)	Zhuang-han yong quantou buduandi chui, hai bu jue de tong. strong-man use fist continuously beat still NEG feel pain 'The strong man continuously beat with his fists, but didn't feel pain.'
	(D) Punctual_Pre+Post (Nontransparent Iteration)	Zhuang-han yong quantou buduandi chui-le 1 fenzhong, strong-man use fist continuously beat-LE 1 minute hai bu jue de tong. still NEG feel pain 'The strong man continuously beat for a minute with his fists, but didn't feel pain.'
s08	(A) Durative_Post (No Iteration)	Muji zai tianli xun-le 10 fenzhong, sihu zai mishi. hen LOC field roam-LE 10 minute seemingly PROG forage 'The hen roamed for 10 minutes in the field, seemingly foraging.'
	(B) Punctual_Post (Nontransparent Iteration)	Muji zai tianli zhuo-le 10 fenzhong, sihu zai mishi. hen LOC field peck-LE 10 minute seemingly PROG forage 'The hen pecked for 10 minutes in the field, seemingly foraging.'
	(C) Punctual_Pre (Nontransparent Iteration)	Muji zai tianli yizhi zhuo, sihu zai mishi. hen LOC field continuously peck seemingly PROG forage 'The hen continuously pecked in the field, seemingly foraging.'
	(D) Punctual_Pre+Post (Nontransparent Iteration)	Muji zai tianli yizhi zhuo-le 10 fenzhong, sihu zai mishi. hen LOC field continuously peck-LE 10 minute seemingly PROG forage 'The hen continuously pecked for 10 minutes in the field, seemingly foraging.'
s09	(A) Durative_Post (No Iteration)	Xiao gou zai dishang shui-le 3 fenzhong, haoxiang hen wuliao. school dog LOC ground sleep-LE 3 minute seem very bored 'The school dog slept for 3 minutes on the ground and seemed very bored.'
	(B) Punctual_Post (Nontransparent Iteration)	Xiao gou zai dishang fan-le 3 fenzhong, haoxiang hen wuliao. school dog LOC ground turn.over-LE 3 minute seem very bored 'The school dog turned over for 3 minutes on the ground and seemed very bored.'
	(C) Punctual_Pre (Nontransparent Iteration)	Xiao gou zai dishang yizhi fan, haoxiang hen wuliao. school dog LOC ground continuously turn-over seem very bored 'The school dog continuously turned over on the ground and seemed very bored.'
	(D) Punctual_Pre+Post (Nontransparent Iteration)	Xiao gou zai dishang yizhi fan-le 3 fenzhong, school dog LOC ground continuously turn.over-LE 3 minute haoxiang hen wuliao. seem very bored 'The school dog continuously turned over for 3 minutes on the ground and seemed very bored.'

s10	(A) Durative_Post (No Iteration)	Riju zhong renzhe yong shou ya-le 2 fenzhong, Japanese drama in ninja use hand press-LE 2 minute heshi de chuang-zhi reng wanhao ruchu. Japanese room DE window-paper still intact like-before 'In the Japanese drama, the ninja pressed for 2 minutes with his hand, yet the window paper of the Japanese room was still intact like before.'
	(B) Punctual_Post (Nontransparent Iteration)	Riju zhong renzhe yong shou chuo-le 2 fenzhong, Japanese drama in ninja use hand poke-LE 2 minute heshi de chuang-zhi reng wanhao ruchu. Japanese room DE window-paper still intact like-new 'In the Japanese drama, the ninja poked for 2 minutes with his hand, yet the window paper of the Japanese room was still intact like before.'
	(C) Punctual_Pre (Nontransparent Iteration)	Riju zhong renzhe yong shou buduandi chuo, Japanese drama in ninja use hand continuously poke heshi de chuang-zhi reng wanhao ruchu. Japanese room DE window-paper still intact like-new In the Japanese drama, the ninja continuously poked with his hand, yet the window paper of the Japanese room was still intact like before.'
	(D) Punctual_Pre+Post (Nontransparent Iteration)	Riju zhong renzhe yong shou buduandi chuo-le 2 fenzhong, Japanese drama in ninja use hand continuously poke-LE 2 minute heshi de chuang-zhi reng wanhao ruchu. Japanese room DE window-paper still intact like-new 'In the Japanese drama, the ninja continuously poked for 2 minutes with his hand, yet the window paper of the Japanese room was still intact like before.'
s11	(A) Durative_Post (No Iteration)	Bisai zhongju shi qiu gun-le 5 miao, haishi diao dao jiewai. game endgame at ball roll-LE 5 second still drop out of bound 'The ball rolled for 5 seconds at the end of the game, still falling out of bounds.'
	(B) Punctual_Post (Nontransparent Iteration)	Bisai zhongju shi qiu dan-le 5 miao, haishi diao dao jiewai. game endgame at ball bounce-LE 5 second still drop out of bound 'The ball bounced for 5 seconds at the end of the game, still falling out of bounds.'
	(C) Punctual_Pre (Nontransparent Iteration)	Bisai zhongju shi qiu buduandi dan, haishi diao dao jiewai. game endgame at ball continuously bounce still drop out of bound 'The ball continuously bounced at the end of the game, still falling out of bounds.'
	(D) Punctual_Pre+Post (Nontransparent Iteration)	Bisai zhongju shi qiu buduandi dan-le 5 miao, game endgame at ball continuously bounce-LE 5 minute haishi diao dao jiewai. still drop out-of-bound 'The ball continuously bounced for 5 seconds at the end of the game, still falling out of bounds.'
s12	(A) Durative_Post (No Iteration)	Dianying zhong nuzhujiao de yanjing bi-le 10 miao, movie in heroine DE eye close-LE 10 second cai jiezhe shuohua. CAI continue speak 'The heroine in the movie closed her eyes for 10 seconds before continuing to speak.'
	(B) Punctual_Post (Nontransparent Iteration)	Dianying zhong nuzhujiao de yanjing zha-le 10 miao, movie in heroine DE eye blink-LE 10 second cai jiezhe shuohua. CAI continue speak 'The heroine in the movie blinked her eyes for 10 seconds before continuing to speak.'

	(C) Punctual_Pre (Nontransparent Iteration)	Dianying zhong nuzhujiao de yanjing buduandi zha, movie in heroine DE eye continuously blink cai jiezhe shuohua. CAI continue speak 'The heroine in the movie continuously blinked her eyes before continuing to speak.'
	(D) Punctual_Pre+Post (Nontransparent Iteration)	Dianying zhong nuzhujiao de yanjing buduandi zha-le 10 miao, movie in heroine DE eye continuously blink-LE 10 second cai jiezhe shuohua. CAI continue speak 'The heroine in the movie continuously blinked her eyes for 10 seconds before continuing to speak.'
s13	(A) Durative_Post (No Iteration)	Zuowan zoulang de dengpao liang-le 2 fenzhong, last night hallway DE light bulb light-LE 2 minute houlai jiu wanquan ximie-le. after then completely extinguish-LE 'The light bulb in the hallway lighted for 2 minutes last night and then extinguished completely.'
	(B) Punctual_Post (Nontransparent Iteration)	Zuowan zoulang de dengpao shan-le 2 fenzhong, last night hallway DE light bulb flicker-LE 2 minute houlai jiu wanquan ximie-le. after then completely extinguish-LE 'The light bulb in the hallway flickered for 2 minutes last night and then extinguished completely.'
	(C) Punctual_Pre (Nontransparent Iteration)	Zuowan zoulang de dengpao yizhi shan, last night hallway DE light bulb continuously flicker houlai jiu wanquan ximie-le. after then completely extinguish-LE 'The light bulb in the hallway continuously flickered last night and then extinguished completely.'
	(D) Punctual_Pre+Post (Nontransparent Iteration)	Zuowan zoulang de dengpao yizhi shan-le 2 fenzhong, last night hallway DE light bulb continuously flicker-LE 2 minute houlai jiu wanquan ximie-le. after then completely extinguish-LE 'The light bulb in the hallway continuously flickered for 2 minutes last night and then extinguished completely.'
s14	(A) Durative_Post (No Iteration)	Wuyan leiiji de yushui liu-le 10 miao, hai di-ge mei-wan. eave accumulate DE rainwater flow-LE 10 second still drip-ADV NEG-finish 'The rainwater accumulated on the eaves flowed for 10 seconds and still kept dripping.'
	(B) Punctual_Post (Nontransparent Iteration)	Wuyan leiiji de yushui di-le 10 miao, hai di-ge mei-wan. eave accumulate DE rainwater drip-LE 10 second still drip-ADV NEG-finish 'The rainwater accumulated on the eaves dripped for 10 seconds and still kept dripping.'
	(C) Punctual_Pre (Nontransparent Iteration)	Wuyan leiiji de yushui yizhi di, hai di-ge mei-wan. eave accumulate DE rainwater continuously drip still drip-ADV NEG-finish 'The rainwater accumulated on the eaves continuously dripped and still kept dripping.'
	(D) Punctual_Pre+Post (Nontransparent Iteration)	Wuyan leiiji de yushui yizhi di-le 10 miao, eave accumulate DE rainwater continuously drip-LE 10 second hai di-ge mei-wan. still drip-ADV NEG-finish 'The rainwater accumulated on the eaves continuously dripped for 10 seconds and still kept dripping.'

s15	(A) Durative_Post (No Iteration)	Binghuan de sizhi tan-le 20 fenzhong, zhengzhuang hen yanzhong. patient DE limb paralyze-LE 20 minute symptom very severe 'The patient's limbs paralyzed for 20 minutes and the symptoms were severe.'
	(B) Punctual_Post (Nontransparent Iteration)	Binghuan de sizhi chan-le 20 fenzhong, zhengzhuang hen yanzhong. patient DE limb tremble-LE 20 minute symptom very severe 'The patient's limbs trembled for 20 minutes and the symptoms were severe.'
	(C) Punctual_Pre (Nontransparent Iteration)	Binghuan de sizhi yizhi chan, zhengzhuang hen yanzhong. patient DE limb continuously tremble symptom very severe 'The patient's limbs continuously trembled and the symptoms were severe.'
	(D) Punctual_Pre+Post (Nontransparent Iteration)	Binghuan de sizhi yizhi chan-le 20 fenzhong, patient DE limb continuously tremble-LE 20 minute zhengzhuang hen yanzhong. symptom very severe 'The patient's limbs continuously trembled for 20 minutes and the symptoms were severe.'
s16	(A) Durative_Post (No Iteration)	Bangongshi de dianhua zhanxian-le 10 fenzhong, office DE phone be occupied-LE 10 minute dou-mei-ren lihui. no one pay.attention 'The office phone was occupied for 10 minutes and no one paid attention.'
	(B) Punctual_Post (Nontransparent Iteration)	Bangongshi de dianhua xiang-le 10 fenzhong, dou-mei-ren lihui. office DE phone ring-LE 10 minute no one pay.attention 'The office phone rang for 10 minutes and no one paid attention.'
	(C) Punctual_Pre (Nontransparent Iteration)	Bangongshi de dianhua yizhi xiang, dou-mei-ren lihui. office DE phone continuously ring no one pay.attention 'The office phone continuously rang and no one paid attention.'
	(D) Punctual_Pre+Post (Nontransparent Iteration)	Bangongshi de dianhua yizhi xiang-le 10 fenzhong, office DE phone continuously ring-LE 10 minute dou-mei-ren lihui. no one pay.attention 'The office phone continuously rang for 10 minutes and no one paid attention.'
s17	(A) Durative_Post (No Iteration)	Zhuo-shang de shouji xianzhi-le 3 fenzhong, table-top DE mobile phone BE.idle-LE 3 minute dou meiyou ren chuli. still NEG person handle 'The mobile phone on the table was idle for 3 minutes, and no one handled it.'
	(B) Punctual_Post (Nontransparent Iteration)	Zhuo-shang de shouji zhen-le 3 fenzhong, table-top DE mobile phone vibrate-LE 3 minute dou meiyou ren chuli. still NEG person handle 'The mobile phone on the table vibrated for 3 minutes, and no one handled it.'
	(C) Punctual_Pre (Nontransparent Iteration)	Zhuo-shang de shouji yizhi zhen, dou meiyou ren chuli. table-top DE mobile phone continuously vibrate still NEG person handle 'The mobile phone on the table continuously vibrated, and no one handled it.'
	(D) Punctual_Pre+Post (Nontransparent Iteration)	Zhuo-shang de shouji yizhi zhen-le 3 fenzhong, table-top DE mobile phone continuously vibrate-LE 3 minute dou meiyou ren chuli. still NEG person handle 'The mobile phone on the table continuously vibrated for 3 minutes, and no one handled it.'

s18	(A) Durative_Post (No Iteration)	Youleyuan li pengpengche ting-le 5 fenzhong, playground LOC bumper-car park-LE 5 minute ye bu-jian gongzuorenyuan you suo fanying. still NEG-see staff have respond 'The bumper cars in the amusement park parked for 5 minutes, yet no staff responded.'
	(B) Punctual_Post (Nontransparent Iteration)	Youleyuan li pengpengche zhuang-le 5 fenzhong, playground LOC bumper-car collide-LE 5 minute ye bu-jian gongzuorenyuan you suo fanying. still NEG-see staff have respond 'The bumper cars in the amusement park collided for 5 minutes, yet no staff responded.'
	(C) Punctual_Pre (Nontransparent Iteration)	Youleyuan li pengpengche buduandi zhuang, playground LOC bumper-car continuously collide ye bu-jian gongzuorenyuan you suo fanying. still NEG-see staff have respond 'The bumper cars in the amusement park continuously collided, yet no staff responded.'
	(D) Punctual_Pre+Post (Nontransparent Iteration)	Youleyuan li pengpengche buduandi zhuang-le 5 fenzhong, playground LOC bumper-car continuously collide-LE 5 minute ye bu-jian gongzuorenyuan you suo fanying. still NEG-see staff have respond 'The bumper cars in the amusement park continuously collided for 5 minutes, yet no staff responded.'
s19	(A) Durative_Post (No Iteration)	Yanshui fengpao xuangua-le 30 fenzhong, canjiazhe xingfen-buyi. Yanshui firework hang-LE 30 minute participant excite-ADV 'The Yanshui fireworks hung for 30 minutes, and the participants were excited.'
	(B) Punctual_Post (Nontransparent Iteration)	Yanshui fengpao zha-le 30 fenzhong, canjiazhe xingfen-buyi. Yanshui firework explode-LE 30 minute participant excite-ADV 'The Yanshui fireworks exploded for 30 minutes, and the participants were excited.'
	(C) Punctual_Pre (Nontransparent Iteration)	Yanshui fengpao buduandi zha, canjiazhe xingfen-buyi. Yanshui firework continuously explode participant excite-ADV 'The Yanshui fireworks continuously exploded, and the participants were excited.'
	(D) Punctual_Pre+Post (Nontransparent Iteration)	Yanshui fengpao buduandi zha-le 30 fenzhong, Yanshui firework continuously explode-LE 30 minute canjiazhe xingfen-buyi. participant excite-ADV 'The Yanshui fireworks continuously exploded for 30 minutes, and the participants were excited.'
s20	(A) Durative_Post (No Iteration)	Youniao de chibang zhang-le 10 fenzhong, haishi meiyong fei-qilai. chick DE wing outstretch-LE 10 minute still NEG flow-COMP 'The chick's wings were outstretched for 10 minutes, but it still didn't fly up.'
	(B) Punctual_Post (Nontransparent Iteration)	Youniao de chibang zhen-le 10 fenzhong, haishi meiyong fei-qilai. chick DE wing flutter-LE 10 minute still NEG flow-COMP 'The chick's wings fluttered for 10 minutes, but it still didn't fly up.'
	(C) Punctual_Pre (Nontransparent Iteration)	Youniao de chibang buduandi zhen haishi meiyong fei-qilai. chick DE wing continuously flutter still NEG flow-COMP 'The chick's wings continuously fluttered, but it still didn't fly up.'
	(D) Punctual_Pre+Post (Nontransparent Iteration)	Youniao de chibang buduandi zhen-le 10 fenzhong, chick DE wing continuously flutter-LE 10 minute haishi meiyong fei-qilai. still NEG flow-COMP 'The chick's wings continuously fluttered for 10 minutes, but it still didn't fly up.'

s21	(A) Durative_Post (No Iteration)	Xiaopengyou zai chuang-shang ku-le 3 fenzhong, buken shanbaganxiu. little child LOC bed-top cry-LE 3 minute NEG-willing give up 'The child cried for 3 minutes on the bed, not willing to give up.'
	(B) Punctual_Post (Nontransparent Iteration)	Xiaopengyou zai chuang-shang tiao-le 3 fenzhong, buken shanbaganxiu. little child LOC bed-top jump-LE 3 minute NEG-willing give up 'The child jumped for 3 minutes on the bed, not willing to give up.'
	(C) Punctual_Pre (Nontransparent Iteration)	Xiaopengyou zai chuang-shang buduandi tiao, buken shanbaganxiu. little child LOC bed-top continuously jump NEG-willing give up 'The child continuously jumped on the bed, not willing to give up.'
	(D) Punctual_Pre+Post (Nontransparent Iteration)	Xiaopengyou zai chuang-shang buduandi tiao-le 3 fenzhong, little child LOC bed-top continuously jump-LE 3 minute bukan shanbaganxiu. NEG-willing give up 'The child continuously jumped for 3 minutes on the bed, not willing to give up.'
s22	(A) Durative_Post (No Iteration)	Dashu yun-le 10 fenzhong, juede shenti bu-tai shufu. uncle faint-LE 10 minute feel body NEG-quite well 'The uncle fainted for 10 minutes, feeling not quite well.'
	(B) Punctual_Post (Nontransparent Iteration)	Dashu ke-le 10 fenzhong, juede shenti bu-tai shufu. uncle cough-LE 10 minute feel body NEG-quite well 'The uncle coughed for 10 minutes, feeling not quite well.'
	(C) Punctual_Pre (Nontransparent Iteration)	Dashu yizhi ke, juede shenti bu-tai shufu. uncle continuously cough feel body NEG-quite well 'The uncle continuously coughed, feeling not quite well.'
	(D) Punctual_Pre+Post (Nontransparent Iteration)	Dashu yizhi ke-le 10 fenzhong, juede shenti bu-tai shufu. uncle continuously cough-LE 10 minute feel body NEG-quite well 'The uncle continuously coughed for 10 minutes, feeling not quite well.'
s23	(A) Durative_Post (No Iteration)	Taiquandao xuanshou dun-le 5 fenzhong, nuli lianxi jiben dongzuo. taekwondo athlete squat-LE 5 minute strive practice basic movements 'The taekwondo athlete squatted for 5 minutes, working hard to practice basic movements.'
	(B) Punctual_Post (Nontransparent Iteration)	Taiquandao xuanshou ti-le 5 fenzhong, nuli lianxi jiben dongzuo. taekwondo athlete kick-LE 5 minute strive practice basic movement 'The taekwondo athlete kicked for 5 minutes, working hard to practice basic movements.'
	(C) Punctual_Pre (Nontransparent Iteration)	Taiquandao xuanshou buduandi ti, nuli lianxi jiben dongzuo. taekwondo athlete continuously kick strive practice basic movements 'The taekwondo athlete continuously kicked, working hard to practice basic movements.'
	(D) Punctual_Pre+Post (Nontransparent Iteration)	Taiquandao xuanshou buduandi ti-le 5 fenzhong, taekwondo athlete continuously kick-LE 5 minute nuli lianxi jiben dongzuo. strive practice basic movements 'The taekwondo athlete continuously kicked for 5 minutes, working hard to practice basic movements.'
s24	(A) Durative_Post (No Iteration)	Moshushi yong shouzhi zhe-le 15 miao, magician use finger cover-LE 15 second cong shouzhong-zhong bianchu yi-duo meiguihua. from palm-middle conjure.up one-CL rose

		'The magician covered for 15 seconds with his fingers, conjuring up a rose from the palm.'
	(B) Punctual_Post (Nontransparent Iteration)	Moshushi yong shouzhi dan-le 15 miao, magician use finger flick-LE 15 second cong shouzhang-zhong bianchu yi-duo meiguihua. from palm-middle conjure.up one-CL rose 'The magician flicked for 15 seconds with his fingers, conjuring up a rose from the palm.'
	(C) Punctual_Pre (Nontransparent Iteration)	Moshushi yong shouzhi buduandi dan, magician use finger continuously flick cong shouzhang-zhong bianchu yi-duo meiguihua. from palm-middle conjure.up one-CL rose 'The magician continuously flicked with his fingers, conjuring up a rose from the palm.'
	(D) Punctual_Pre+Post (Nontransparent Iteration)	Moshushi yong shouzhi buduandi dan-le 15 miao, magician use finger continuously flick-LE 15 second cong shouzhang-zhong bianchu yi-duo meiguihua. from palm-middle conjure.up one-CL rose 'The magician continuously flicked for 15 seconds with his fingers, conjuring up a rose from the palm.'
s25	(A) Durative_Post (No Iteration)	Baomu qingqingde chang-le 20 fenzhong, nanny gently sing-LE 20 minute cai rang xiaoying'er guaiguai shuizhe. CAI let baby obediently fall asleep 'The nanny gently sang for 20 minutes and let the baby fell asleep obediently.'
	(B) Punctual_Post (Nontransparent Iteration)	Baomu qingqingde pai-le 20 fenzhong, nanny gently pat-LE 20 minute cai rang xiaoying'er guaiguai shuizhe. CAI let baby obediently fall asleep 'The nanny gently patted for 20 minutes and let the baby fell asleep obediently.'
	(C) Punctual_Pre (Nontransparent Iteration)	Baomu qingqingde yizhi pai, cai rang xiaoying'er guaiguai shuizhe nanny gently continuously pat CAI let baby fall obediently asleep 'The nanny gently and continuously patted and let the baby fell asleep obediently.'
	(D) Punctual_Pre+Post (Nontransparent Iteration)	Baomu qingqingde yizhi pai-le. 20 fenzhong, nanny gently continuously pat-LE 20 minute cai rang xiaoying'er guaiguai shuizhe. CAI let baby obediently fall asleep 'The nanny gently and continuously patted for 20 minutes and let the baby fell asleep obediently.'
s26	(A) Durative_Post (No Iteration)	Dianshi yingmu guzhang-le 10 miao, guo yi zhenzi cai huifu zhengchang. TV screen fault-LE 10 second pass a while CAI return normal 'The TV screen faulted for 10 seconds and returned to normal after a while.'
	(B) Punctual_Post (Nontransparent Iteration)	Dianshi yingmu shan-le 10 miao, guo yi zhenzi cai huifu zhengchang. TV screen flicker-LE 10 second pass a while CAI return normal 'The TV screen flickered for 10 seconds and returned to normal after a while.'
	(C) Punctual_Pre (Nontransparent Iteration)	Dianshi yingmu buduandi shan, guo yi zhenzi cai huifu zhengchang. TV screen continuously flicker pass a while CAI return normal 'The TV screen continuously flickered and returned to normal after a while.'

	(D) Punctual_Pre+Post (Nontransparent Iteration)	Dianshi yingmu buduandi shan-le 10 miao, TV screen continuously flicker-LE 10 second guo yi zhenzi cai huifu zhengchang. after a while CAI return normal 'The TV screen continuously flickered for 10 seconds and returned to normal after a while.'
s27	(A) Durative_Post (No Iteration)	Yushi de shui liu-le 10 fenzhong, houlai cai bathroom DE water flow-LE 10 minute afterward CAI bei faxian shuilongtou meiyou guanjin. BEI discover faucet NEG turn.off-tight 'The water of the bathroom flowed for 10 minutes and it was discovered afterward that the faucet was not turned off tightly.'
	(B) Punctual_Post (Nontransparent Iteration)	Yushi de shui di-le 10 fenzhong, houlai cai bathroom DE water drip-LE 10 minute afterward CAI bei faxian shuilongtou meiyou guanjin. BEI discover faucet NEG turn.off-tight 'The water of the bathroom dripped for 10 minutes and it was discovered afterward that the faucet was not turned off tightly.'
	(C) Punctual_Pre (Nontransparent Iteration)	Yushi de shui yizhi di, houlai cai bathroom DE water continuously drip afterward CAI bei faxian shuilongtou meiyou guanjin. BEI discover faucet NEG turn.off-tight 'The water of the bathroom continuously dripped and it was discovered afterward that the faucet was not turned off tightly.'
	(D) Punctual_Pre+Post (Nontransparent Iteration)	Yushi de shui yizhi di-le 10 fenzhong, houlai cai bathroom DE water continuously drip-LE 10 minute afterward CAI bei faxian shuilongtou meiyou guanjin. BEI discover facet NEG turn.off-tight 'The water of the bathroom continuously dripped for 10 minutes and it was discovered afterward that the faucet was not turned off tightly.'
s28	(A) Durative_Post (No Iteration)	Xiao nuhai yanjing bi-le 10 miaozhong, dagai shi jue de dengguang little girl eye close-LE 10 second probably feel light hen ciyan. very dazzling 'The little girl closed her eyes for 10 seconds, probably finding the light dazzling.'
	(B) Punctual_Post (Nontransparent Iteration)	Xiao nuhai yanjing zha-le 10 miaozhong, dagai shi jue de dengguang little girl eye blink-LE 10 second probably feel light hen ciyan. very dazzling 'The little girl blinked her eyes for 10 seconds, probably finding the light dazzling.'
	(C) Punctual_Pre (Nontransparent Iteration)	Xiao nuhai yanjing buduandi zha, dagai shi jue de dengguang hen ciyan. little girl eye continuously blink probably feel light very dazzling 'The little girl continuously blinked her eyes, probably finding the light dazzling.'
	(D) Punctual_Pre+Post (Nontransparent Iteration)	Xiao nuhai yanjing buduandi zha-le 10 miaozhong, dagai shi jue de little girl eye continuously blink-LE 10 second probably feel dengguang hen ciyan. light very dazzling 'The little girl continuously blinked her eyes for 10 seconds, probably finding the light dazzling.'
s29	(A) Durative_Post (No Iteration)	Gaoerfu qiu zai changshang fei-le 15 miaozhong, golf ball LOC field fly-LE 15 second zuizhong haishi mei jin dong. finally still NEG enter hole 'The golf ball flew for 15 seconds on the field, and finally missed the hole still.'

	(B) Punctual_Post (Nontransparent Iteration)	Gaoerfu qiu zai changshang dan-le 15 miaozhong, colf ball LOC field bounce-LE 15 second zuizhong haishi mei jin dong. finally still NEG enter hole 'The golf ball bounced for 15 seconds on the field, and finally missed the hole still.'
	(C) Punctual_Pre (Nontransparent Iteration)	Gaoerfu qiu zai changshang yizhi dan, golf ball LOC field continuously bounce zuizhong haishi mei jin dong. finally still NEG enter hole 'The golf ball continuously bounced on the field, and finally missed the hole still.'
	(D) Punctual_Pre+Post (Nontransparent Iteration)	Gaoerfu qiu zai changshang yizhi dan-le 15 miaozhong, golf ball LOC field continuously bounce-LE 15 second zuizhong haishi mei jin dong. finally still NEG enter hole 'The golf ball continuously bounced for 15 seconds on the field, and finally missed the hole still.'
	(A) Durative_Post (No Iteration)	Bingren de shuang shou ma-le 10 fenzhong, yuanyin bumeng. patient DE both hand numb-LE 10 minute reason unknown 'The patient's hands went numb for 10 minutes for unknown reasons.'
s30	(B) Punctual_Post (Nontransparent Iteration)	Bingren de shuang shou dou-le 10 fenzhong, yuanyin bumeng. patient DE both hand tremble-LE 10 minute reason unknown 'The patient's hands trembled for 10 minutes for unknown reasons.'
	(C) Punctual_Pre (Nontransparent Iteration)	Bingren de shuang shou yizhi dou, yuanyin bumeng. patient DE both hand continuously tremble reason unknown 'The patient's hands continuously trembled for unknown reasons.'
	(D) Punctual_Pre+Post (Nontransparent Iteration)	Bingren de shuang shou yizhi dou-le 10 fenzhong, patient DE both hand continuously tremble-LE 10 minute yuanyin bumeng. reason unknown 'The patient's hands continuously trembled for 10 minutes for unknown reasons.'

Appendix A-2. Stimuli – Nonsensical sentences

Set#	Condition	Target Sentence
f31	(E) Achiev_DurAdv (Nonsensical)	Yundongyuan zongjuesai ying-le 10 fenzhong, xiande hen deyi. athlete final game win-LE 10 minute appear very proud 'The athlete won the final game for 10 minutes, appearing very proud.'
	(F) State_DurAdv (Nonsensical)	Yundongyuan zongjuesai gao-le 10 fenzhong, xiande hen deyi. athlete final game be.tall-LE 10 minute appear very proud 'The athlete was tall for 10 minutes, appearing very proud.'
	(G) SemViolation (Nonsensical)	Jiaokeshu zai caochang pao-le yi-ge shangwu, xiande hen deyi. textbook LOC playground run-LE one-CL morning appear very proud 'The textbook ran in the playground for the whole morning, appearing very proud.'
f32	(E) Achiev_DurAdv (Nonsensical)	Xuesheng kaoshi shi wangji-le 1 xiaoshi. student exam during forget-LE 1 hour 'The student forgot for an hour during the exam.'

	(F) State_DurAdv (Nonsensical)	Xuesheng kaoshi shi congming-le 1 xiaoshi. student exam during smart-LE 1 hour 'The student was smart for an hour during the exam.'
	(G) SemViolation (Nonsensical)	Qianbi shui-le 1 xiaoshi cai youyou zhuanxing. pencil sleep-LE 1 hour CAI leisurely wake up 'The pencil slept for an hour and leisurely waking up.'
f33	(E) Achiev_DurAdv (Nonsensical)	Dahua ba beizi shuaipo-le 2 xiaoshi, rede laoshi bu gaoxing. Dahua BA cup break-LE 2 hour let teacher NEG happy Dahua broke the cup for 2 hours, making the teacher unhappy.'
	(F) State_DurAdv (Nonsensical)	Dahua zai 2 xiaoshi nei jiao'ao-le, rede laoshi bu gaoxing. Dahua in 2 hour within proud-LE let teacher NEG happy 'Dahua was proud in 2 hours, making the teacher unhappy.'
	(G) SemViolation (Nonsensical)	Beizi zai zhuo-shang zou-lai-zou-qu, rede laoshi bu gaoxing. cup LOC table-top walk around let teacher NEG happy 'The cup walked around on the table, making the teacher unhappy.'
f34	(E) Achiev_DurAdv (Nonsensical)	Linju de xiaohai kanjian-le ta baba 5 fenzhong, juede hen kaixin. neighbor DE child see-LE 3SG father 5 minute feel very happy 'The neighbor's kid saw his father for 5 minutes, feeling very happy.'
	(F) State_DurAdv (Nonsensical)	Linju de xiaohai xiang-le ta baba 5 fenzhong, juede hen kaixin. neighbor DE child resemble-LE 3SG father 5 minute feel very happy 'The neighbor's kid resembled his father for 5 minutes, feeling very happy.'
	(G) SemViolation (Nonsensical)	Linju de lese wan-le hao yizhenzi, juede hen kaixin. neighbor DE garbage play-LE quite a while feel very happy 'The neighbor's garbage played for quite a while, feeling very happy.'
f35	(E) Achiev_DurAdv (Nonsensical)	Jingfang faxian-le kekao xiansuo 10 fenzhong, you kuaisu poan police find-LE reliable clue 10 minute have quick solve.case de keneng. DE possibility 'The police found reliable clues for 10 minutes, and it was possible to solve the case quickly.'
	(F) State_DurAdv (Nonsensical)	Jingfang zai 10 fenzhong nei rexin-le, you kuaisu poan police LOC 10 minute within be.enthusiastic-LE have quick solve.case de keneng. DE possibility 'The police was enthusiastic within 10 minutes, and it was possible to solve the case quickly.'
	(G) SemViolation (Nonsensical)	Xiansuo dun zai yuandi yidong-ye-budong, you kuaisu poan clue squat LOC in-situ NEG move any have quick solve.case de keneng. DE possibility 'The clues squatted in-situ without moving, and it was possible to solve the case quickly.'
f36	(E) Achiev_DurAdv (Nonsensical)	Zhe zhou shangke shi xin lai de xuesheng mingbai-le this week class during new come DE student understand-LE 1-ge xiaoshi. 1-CL hour 'The new students understood for an hour in the class this week.'
	(F) State_DurAdv (Nonsensical)	Zhe zhou shangke shi xin lai de xuesheng zai 1-ge xiaoshi nei. this week class during new come DE student in 1-CL hour within yonggan-le. be.brave-LE 'The new students was brave within an hour in the class this week.'

	(G) SemViolation (Nonsensical)	Zhe zhou shangke de jiaoshi shui-le zhengzheng yi-tian. this week course DE classroom sleep-LE whole one-day 'The classroom of the course for this week slept all day.'
f37	(E) Achiev_DurAdv (Nonsensical)	Shequ jingwei rende-le heiguanmalu 5 fenzhong. community guard recognize-LE Malayan Night-Heron 5 minute 'The community guard recognized the Malayan Night-Heron (Gorsachius melanolophus) for 5 minutes.'
	(F) State_DurAdv (Nonsensical)	Shequ jingwei zai 5 fenzhong nei haoke-le. community guard in 5 minute within be.hospitable-LE 'The community guard was hospitable within 5 minutes.'
	(G) SemViolation (Nonsensical)	Shequ dalou wucan chi-le yi-ge ban xiaoshi. community building lunch eat-LE one-CL half hour 'The community building had lunch for an hour and a half.'
f38	(E) Achiev_DurAdv (Nonsensical)	Huajia ba xin huace wancheng-le 3 fenzhong, feichang jingren. painter BA new album complete-LE 3 minute very amazing 'The painter completed the new album for 3 minutes, which is very amazing.'
	(F) State_DurAdv (Nonsensical)	Huajia de xin huace hou-le 3 fenzhong, feichang jingren. painter DE new album be.thick-LE 3 minute very amazing 'The painter's new album thickened for 3 minutes, which is very amazing.'
	(G) SemViolation (Nonsensical)	Huajia de xin huace zai caochang zou-le 3 fenzhong. painter DE new album LOC playground walk-LE 3 minute 'The painter's new album walked for 3 minutes in the playground.'
f39	(E) Achiev_DurAdv (Nonsensical)	Banshang de meiguo qiao sheng ba zhongwen xuehui-le class DE American overseas Chinese student BA Chinese learned-LE 10 miaozechong. 10 second 'The American overseas Chinese student in the class learned Chinese for 10 seconds.'
	(F) State_DurAdv (Nonsensical)	Banshang de meiguo qiao sheng zai 10 miaozechong nei class DE American overseas Chinese student LOC 10 second within yongyou lan yanjing. possess blue eye 'The American overseas Chinese student in the class possessed blue eyes for 10 seconds.'
	(G) SemViolation (Nonsensical)	Jiaoshi de ke-zhuo-yi mengyou-le zhengge hanjia. classroom DE desk-and-chair sleepwalk-LE whole winter vacation The desks and chairs of the classroom sleepwalked for the whole winter vacation.'
f40	(E) Achiev_DurAdv (Nonsensical)	Lao wang letou zhong-le 5 fenzhong, xingfen-de budeliao. Mr. Wang lottery win-LE 5 minute excite-ADV extremely 'Mr. Wang won the lottery for 5 minutes and was extremely excited.'
	(F) State_DurAdv (Nonsensical)	Lao wang de shen-xing xiuchang-le 5 fenzhong, xingfen-de budele. Mr. Wang DE body-shape be.slender-LE 5 minute excite-ADV extremely 'Mr. Wang's body shape was slender for 5 minutes and was extremely excited.'
	(G) SemViolation (Nonsensical)	Weishengzhi ku-le hao jiu, haishi jue de ganshang. tissue cry-LE very long still feel sentimental 'The tissues cried for so long but still felt sentimental.'
f41	(E) Achiev_DurAdv (Nonsensical)	Jingguan jiang dupin xunhuo-le 20 fenzhong, you zhongda tupo. officer JIANG drug find-LE 20 minute have significant breakthrough 'The officers found the drugs for 20 minutes, having a significant breakthrough.'

	(F) State_DurAdv (Nonsensical)	Jingguan ai-le 20 fenzhong, juede hen qiguai. officer be.short-LE 20 minute feel very strange 'The officer was short for 20 minutes, feeling that it was strange.'
	(G) SemViolation (Nonsensical)	Dupin gongji-le jingguan, taidu qiangying. drug attack-LE officer attitude tough 'The drugs attacked the officers with a tough attitude.'
f42	(E) Achiev_DurAdv (Nonsensical)	Xiao nanhai renchu-le pengyou 30 fenzhong, xichuwangwai. little boy recognize-LE friend 30 minute be.overjoyed 'The boy recognized his friend for 30 minutes and was overjoyed.'
	(F) State_DurAdv (Nonsensical)	Xiao nanhai zai 30 fenzhong nei yueguan-le, xichuwangwai. little boy LOC 30 minute within be.optimistic-LE be.overjoyed 'The boy was optimistic within 30 minutes and was overjoyed.'
	(G) SemViolation (Nonsensical)	Xichenqi dui-zhe saozhou daxiao, feichang kuangwang. vacuum direct.at-ZHE broom laugh-loud very arrogant 'The vacuum cleaner laughed at the broom, being very arrogant.'
f43	(E) Achiev_DurAdv (Nonsensical)	Shang jie weiyuanzhang zheci xuanju shu-le 3 fenzhong. last term chairman this time election lose-LE 3 minute 'The last chairman lost this election for 3 minutes.'
	(F) State_DurAdv (Nonsensical)	Shang jie weiyuanzhang zheci xuanju keqi-le 3 fenzhong. last term chairman this time election be.polite-LE 3 minute 'The last chairman was polite in this election for 3 minutes.'
	(G) SemViolation (Nonsensical)	Yanjing zai zhuo-shang butingdi aihao. glass LOC table-top nonstop wail 'The glasses wailed nonstop on the table.'
f44	(E) Achiev_DurAdv (Nonsensical)	Nuyanyuan huojiang-le 10 fenzhong, huode zhongren de zanmei. actress win.award-LE 10 minute obtain people DE praise 'The actress won the award for 10 minutes and obtained people's praise.'
	(F) State_DurAdv (Nonsensical)	Nuyanyuan piaoliang-le 10 fenzhong, huode zhongren de zanmei. actress be.pretty-LE 10 minute obtain people. DE praise 'The actress was pretty for 10 minutes and obtained people's praise.'
	(G) SemViolation (Nonsensical)	Maikefeng jia zai taishang pao-le 10 fenzhong, huode zhongren de zanmei. microphone rack LOC stage run-LE 10 minute obtain people DE praise 'The microphone rack ran on the stage for 10 minutes and obtained people's praise.'
f45	(E) Achiev_DurAdv (Nonsensical)	Luoshui de nanzi huojiu-le 20 fenzhong, kanqilai jinghunweiding. fall-water DE man be.rescued-LE 20 minute look terrified 'The man who fell into water was rescued for 20 minutes and looked terrified.'
	(F) State_DurAdv (Nonsensical)	Luoshui de nanzi zai. 20 fenzhong nei danxiao-le, kanqilai jinghunweiding fall-water DE man LOC 20 minute within be.timid-LE look terrified 'The man who fell into water was timid within 20 minutes and looked terrified.'
	(G) SemViolation (Nonsensical)	Jiuping gui zai zuoyi xujiu, kanqilai shensejinghuang. wine bottle kneel LOC seat long time look panicked 'The wine bottle knelt on the seat for a long time, looking panicked.'
f46	(E) Achiev_DurAdv (Nonsensical)	Daoyan shiqu-le fuqin 30 fenzhong, zhengge ren shenseqiaocui. director lose-LE father 30 minute whole person be.haggard 'The director lost his father for 30 minutes, being haggard.'
	(F) State_DurAdv (Nonsensical)	Daoyan zai 30 fenzhong nei pang-le, zhengge ren shenseqiaocui. director LOC 30 minute withing be.fat-LE whole person be.haggard 'The director was fat for 30 minutes, being haggard.'

	(G) SemViolation (Nonsensical)	Bingxiang duo-le yi zheng tian, zhengge ren shenseqiaocui. refrigerator hide-LE one whole day whole person be.haggard 'The refrigerator hid for a whole day, being haggard.'
f47	(E) Achiev_DurAdv (Nonsensical)	Famingjia kefu-kunnan-le 10 fenzhong, zhongyu you xinde jinzhan. inventor overcome-difficulty-LE 10 minute finally have new progress 'The inventor overcame the difficulty for 10 minutes, finally having new progress.'
	(F) State_DurAdv (Nonsensical)	Famingjia zai 10 fenzhong nei weida-le. inventor LOC 10 minute within be.great-LE 'The inventors were great in 10 minutes.'
	(G) SemViolation (Nonsensical)	Yizi sikao-le yige xiaowu, zhongyu you xinde jinzhan. chair ponder-LE one-CL afternoon finally have new progress 'The chair pondered for the whole afternoon, finally having new progress.'
f48	(E) Achiev_DurAdv (Nonsensical)	Linju taitai zhaodao-le mitu xiaomao 20 fenzhong. neighbor wife find-LE lost kitten 20 minute 'The neighbor's wife found the lost kitten for 20 minutes.'
	(F) State_DurAdv (Nonsensical)	Linju taitai dunhou-le 20 fenzhong. neighbor wife be.honest-LE 20 minute 'The neighbor's wife was honest for 20 minutes.'
	(G) SemViolation (Nonsensical)	Jiangyongao zai chufang zhan-le 20 fenzhong. soy sauce paste LOC kitchen stand-LE 20 minute 'The soy sauce paste stood in the kitchen for 20 minutes.'
f49	(E) Achiev_DurAdv (Nonsensical)	Xiaoxuesheng biye-le 15 fenzhong, leiliumanmian. elementary student graduate-LE 15 minute burst into tear 'The elementary student graduated for 15 minutes, bursting into tears.'
	(F) State_DurAdv (Nonsensical)	Xiaoxuesheng ai-le 15 fenzhong, lei liu manmian. elementary student be.short-LE 15 minute burst into tear 'The elementary student was short for 15 minutes, bursting into tears.'
	(G) SemViolation (Nonsensical)	Biyezhengshu ku-le 15 fenzhong, lei liu manmian. graduation certificate cry-LE 15 minute burst into tear 'The graduation certificate cried for 15 minutes, bursting into tears.'
f50	(E) Achiev_DurAdv (Nonsensical)	Chengbanren jiang zheci de gongzuo wancheng-le 5 fenzhong, manager JIANG this time DE work complete-LE 5 minute cai fangxin likai. CAI at.ease leave 'The manager completed the work for 5 minutes before leaving at ease.'
	(F) State_DurAdv (Nonsensical)	Zheci de gongzuo zongzhao leguan-le 5 fenzhong, cai fangxin likai. this time DE work manager be.optimistic-LE 5 minute CAI at.east leave 'The executive of this work was optimistic for 5 minutes before leaving at ease.'
	(G) SemViolation (Nonsensical)	Waitao zai bangongshi jiancha-le yi zheng tian, cai fangxin likai. coat LOC office check-LE one whole day CAI at.east leave 'The coat checked in the office for the whole day before leaving at ease.'

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Yao-Ying Lai has been investigating the neurocognitive mechanisms of real-time language processing on an interdisciplinary approach, using various behavioral and neurological techniques. Her research seeks to understand how the human cognitive system works to obtain the meaning of linguistic expressions in context, grounded on theories of linguistics and cognitive science. The research themes center on lexical-conceptual representation, meaning composition and underspecification, contextualization of sentential meaning, as well as the architecture of conceptual structure that supports human language processing. Recent investigation focuses on the impact of social cognition on language comprehension and individual variability in online meaning computation.

Huei-ling Lai's research centers on the semantic structure of language, focusing on lexical semantics and constructions, grammaticalization, metaphor and metonymy. Methods of cognitive semantics and construction grammar are mainly employed to understand the architecture of linguistic meaning. Her work implements corpus-based investigation to reveal the form and function of semantic devices in Mandarin Chinese and Hakka. She has recently established the Taiwan Hakka Corpus, in which written and spoken data are digitalized, allowing researchers to explore lexical use and constructions of Taiwan Hakka varieties. In addition, her work incorporates computational modeling using machine learning to examine polysemy and sense disambiguation.